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CS402 GRAND QUIZ

Sr.No	MCQS	ANSWER
1.	Which of the following steps replaces multiple incoming transition edges with a single one proving Kleene's theorem part II?	Step 3
2.	Let FA1 accepts many strings and FA2 accepts no strings, then FA1+FA2 will be equal to:	FA1
3.	Let L be the language of all strings, defined over ∑ =(0,1), ending in 10. Which of the following strings are indistinguishable w.r.t L with z being 0?	(010,101)
4.	If r1=(aa+bb)and r2=(a+b) then the language (aa+bb)(a+b) will bhe generated by_____	(r)(r2)
5.	Introducing a new start state in the case of multiple states is step no._____of proving Kleen's theorem part II.	1
6.	The language having even numbers of a's and even number of b's defined over S={a,b} is called _____	EVEN-EVEN
7.	In NFA having no transition at a certain state, FA can be built by introducing :	Empty state
8.	For every three regular expression R, S, and T. the language denoted by R(S U T) and (RS)U(RT) are the _____	Same
9.	Which of the following string belongs to the language of the regular expression (aa*b)?	aabaab
10.	If L1' and L2' are regular languages, then L1, L2 will be	Regular
11.	Suppose the language L1 has 2 L2 has 2 states. If we have a machine M that accepts L1∩L2. Then the total number of states in M is equal to_____	4

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12.	If L is a regular language, then (L')' U L will be:	L
13.	In Mealy machine, the output depends on_____	Present state and Present output
14.	Strings x,y,z belongs to $\sum^*$ such that $xz \in L$ but $yz \notin L$ where $L \subseteq \sum^*$ are:	Distinguishable
15.	Melay machine to increase the output string in magnitude by 1 is called:	Incrementing machine
16.	Suppose we have FA3 (which is to FA1+FA2) then the final state of FA3 will be declared final if:	It corresponds to any of the final states of FA1 or FA2
17.	If we have the finite language and the number of states in the FA is n then the maximum number of letter in each word of the language that will be accepted by the given FA will be :	n+1
18.	Which of the following state is introduced while developing NFA for the closure of FA?	An initial state which should be final as well
19.	Length of EVEN-EVEN language is_____	Even
20.	If FA1 corresponds to (a+b)* then FA1 must accept_____ strings/strings.	Every
21.	In FA, initial stage is represented by :	Drawing an arrowhead before the state
22.	Which one of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?	Mealy machine
23.	Length of machine "AbBAbcd" defined over $\sum$=(Ab,B,c,d) is_____.	Five
24.	An FA is a collection of:	Finite states , finite transition and finite input

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		letter
25.	Given the language L={ab, aa, baa}, which of the following strings are in L*? i) abaabaaabaa ii) aaaabaaaa iii) baaaaabaaaab iv) baaaaabaa	1,2 & 4
26.	In the context to make NFA for the concentration of FA1 and FA2 (FA1 accepting null string) which of the following option is correct?	Final states in both FAs
27.	Every ___ is a ___ as well, but the converse may not be true.	FA, TG
28.	NFA with null string has ____ initial state(s).	One
29.	While finding RE corresponding to a TG, we connect the new start state with the old start state by____ transition.	Null
30.	If S=(x). then S* will be_____.	{^,x,xx,xxx,xxxx….}
31.	The minimum length of string (except null string) of a language that starts and ends in the same letter will be:	2
32.	If S={ab, bb} , then S* will not contain_____.	bbba
33.	Which of the following machine has only one initial state?	Moore machine
34.	Which of the following diagram is very rigid in order to express any language?	FA
35.	Let L be the language of all strings, defined over ∑ =(0,1), ending in 111. Which of the following strings are indistinguishable w.r.t L with z being 11?	111,101
36.	Mealy machine can have _____ final states.	Zero
37.	In Moore machine , output is produced over the change of:	States
38.	Lets we have two regular expressions R1 ={xx+yy}and R2={x+y} . Which one of the following is the correct regular expression for the union of R1 and R2?	(xx+yy)+ (x+y)

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39.	FA corresponding to an NFA can be built by introducing a state corresponding to the combination of states, for a letter having	No transition at certain state
40.	The situation there is no way to leave after entry is called	Davey john locker
41.	Which one of the following word is not accepted by the given regular expression?	Abbaab
42.	According to the theory of automata, there are _____ types of languages.	Two
43.	Which of the following word is not accepted by the given regular expression?	Baabaabba
44.	Regular languages are closed under the following operations.	Union, Concentration , Closure
45.	There can be more than ____ FA for a certain language but for ____FA there is only one language associated with it.	One, one
46.	There is no compulsion that each state must have an on outdoing edge for every input variable in:	Transition graph
47.	FA is also called_____.	DFA
48.	If r1 and r2 are regular expressions then (r1*r2) is_____	RE (Regular expression)
49.	Which of the following is the minimal number of states for a finite automaton accepting the language of all strings defined over any alphabet set?	1
50.	Keeping view language of all strings ending with 'a' for which symbol we will take a loop on the final state of its transition diagram?	A
51.	Which of the following statements is true about NFA with Null String?	Transition of null string is allowed in any stage
52.	Which one of the following diagrams expresses languages more simply?	GTG
53.	The language of all strings defined over alphabet	a and ^

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	set={a,b} that does not end with 'a' actually end with:	
54.	Let S={aa,bb}, then S* will have the _____ string.	^
55.	Formal is also known as_____.	Syntactic language
56.	There may be more than one transition for a certain letter on a state in:	Non-Deterministic finite automata
57.	A loop at a state is supposed to be_____transition while converting Moore machine into an equivalent Maley machine.	Incoming
58.	FA of EVEN language shows null string when_____.	initial state is final as well
59.	Which of the following statement is true about GTG?	Transitions are based on regular expressions
60.	Which of the following machine has only one initial state and no final state?	Moore machine
61.	In GTG, there can be more than one :	Start state and final state
62.	GTG for the expression (aa+aba)* may have minimum number of states.	1
63.	In regular expression, the operator '*' stands for.	Iteration
64.	If we have only one state having no transition for input letters, then it is an example of:	NFA
65.	If A and B are regular languages, !(A'U B') is:	Regular
66.	A _____ with "n" states must accept least one string of length greater than "n".	DFA
67.	If r1 is a regular expression then (r1)* is _____.	Also a regular expression
68.	Which of the following is the bypass and state elimination step in the context of Kleene's theorem part II proof?	4
69.	Which of the following is free of non-	FA

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	determinism?	
70.	There is no question of accepting any language in:	Moore machine
71.	A string will be accepted by an NFA if there exist_____one successful path.	atleast
72.	Kleene's theorem part I expresses the relationship between_____	FA and TG
73.	Keeping in view the discussion by Martin, how any states are required to recognize the language of all strings of length 2 or more defined over ∑={a,b} with 'b' being the second letter from right?	7
74.	FA and _____ are same excepts that _____has unique symbol for each transition.	NFA,FA
75.	Subtraction of binary number is possible through.	Both complementing and incrementing machine
76.	Null strings can be specified on edges in:	Transition graph
77.	What is false about the PLAINDROME LANGUAGE?	FA can be build for it
78.	While finding RE corresponding to a TG, if TG has more than one start state then.	Introduce the new start state
79.	All possible combination of string of a language including null string is referred as.	Kleene star closure of a language
80.	N! will be equal to:	n*(n-1)!
81.	Every NFA can be considered to be a____ as well, but the converse may not be true.	TG
82.	In proving Kleene theorem II , if three statuses are connected then middle state is removed bt connecting first and third and writing corresponding RE in:	Concentration
83.	In _____ there must be transition for all the letters of the string.	FA

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84.	For a given Moore Machine , the input string is '101010' , thus the output string would be of length:	Length of input string+1
85.	The FA can be drawn for the regular expression (a+b)* with minimum_____ state(s).	1
86.	Which of the following does not contribute while finding out the length of strings?	^
87.	The language of all strings defined over alphabet set ={x,y} that ends with same letters will have the maximum length of:	Infinite
88.	Considering FA1 and FA2 having 2 states each. Now FA1+FA2 can have maximum_____ number of states.	More than 3
89.	Which one of the following is RE for the language defined over ∑={a,b} having all the words starting with a?	a(a+b)*
90.	An _____ can be considered to be an intermediate structure between finite automaton and transition graph.	NFA
91.	In order to make NFA for the union of FA1 and FA2, the new initial state should be liked to:	Initial states of both FAs
92.	We cannot construct an NFA for the language of _____ defined over alphabet set{a,b}.	Palindromes
93.	The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by the …… production trees	More then one / Different
94.	What do automata mean?	something that works automatically / "something are done automatically
95.	According to theory of automata there	2
96.	The Minimum length of the string (except bull string) of a language that starts and ends in the same letters will be:	1

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97.	If S= {ab,bb} then S* will not contain_____	Bbba
98.	Which of the following machines has only one initial state and no final state?	More machine
99.	Which of the following diagram is very rigid in order to express any language ?	FA
100.	If S={a},then S+will be_____	{a,aa ,aaa , aaaa…}
101.	Let L be the language of the all string defined over over∑ ¿{0,1} ending in10. Which of the following string are indistinguishable with respect to L with z being 11?	010,101
102.	Melay machine can have _____final state	Zero
103.	Given the Language L = {ab, aa,baa},which of the following string are in L*? 1) Abaabaaaabaa 2) Aaaabaaaa 3) Baaaaabaaaaab 4) baaaaabaa	1, 2 and 4
104.	If L1 and L2 are regular Language L1∩(L2∪L1) will be	Regular
105.	In Mealy Machine the out put depends on_____	Present State and present output
106.	There is no question of accepting any language in.	Moore Machine
107.	The state where there is no way to leave after entry is called___	Davey john Locker
108.	FA corresponding to an NFA can be built by introducing an empty state for a letter having	No transaction at certain state
109.	Which of the following diagram express language more simply?	GTG
110.	Automata is the plural of_____	Automation
111.	If A and B are regular Language !(A∪B) is	Regular
112.	In NFA having no transition at certain state FA can be built by introducing.	Empty state

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113.	Consuming FA1 and FA2 having 2 STATES each. Now FA1+FA2 can have maximum_____number of state	2 3 More then 3 None of the given
114.	In an FA when there is no path starting from initial state and ending state in final state then that FA_____	Accept all string
115.	According to theory of automation there are_____type if language	Two
116.	In Moore machine if the length of input string is 9 then the length of output string will be.	10
117.	When ODD language is expressed by an FA,then it will have minimum ____ state	One
118.	[(a+b)(a+b)]*.given RE contact generate the string_____	Bbbbbb
119.	Which of the following state is true about GTG?	Transection are based on regular expression
120.	Every_____is a _____ as will, but the converts may not be true.	FA,TG
121.	Which of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?	Mealy Machine
122.	The recursive method for defining a language has ____ steps	Three
123.	Consider the following RE. A(a+b)b* All of the following word are accepted except_____	Aa
124.	Which of the following regular expression represented same language? 1. (a+ab)* 2. 2(ba+a)* 3. A*(aa*b)*	1 and 2

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	4. (a*b)*	
125.	For every there regular expression R,S and T the Language denoted by R(S∪T) and (RS)∪(RT) are the	Same
126.	Alphabet S={a,bc,cc} has____number of letters.	Three
127.	An____can be considered to be an intermediate structure between Finite automation and Transition Graph.	NFA
128.	Two FAs are said to be equivalent if they_____	Accept same language
129.	There may be more then one transition for a certain letter on a state in	Finite Automata
130.	_____can also help in proving Kleene Theorem III.	NFA
131.	Kleene Theorem Part II expression the relationship between____	TG and RE
132.	FA corresponding to an NFA can be built by introducing an empty state for a letter having.	No transition at certain state
133.	FA is also called	DFA
134.	If two Res generated same language then these Res are called ___	Equivalent RE
135.	We cannot an NFA for the language of ____ defend over alphabet set{a,b}	Palindromes
136.	Kleene Theorem Part II expression the relationship between____	RE and FA
137.	Let FA3 an FA corresponding to FA1 FA2 then the initial state of FA3 must correspond to the initial state of	FA1 only
138.	Every FA should be___	Deterministic
139.	The minimum length of string (except null string) of a language that starts and ends in the different letter will be:	1
140.	Which of the following will be the final state of FA3 obtained from the union of FA1 and FA2?	Final states of FA1 or FA2
141.	In concatenation we accept the initial state of FA2 automatically after the final state of FA1	We need just one initial

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	because of:	state
142.	Let FA1 accepts many strings and FA2 accepts none then FA1+FA2 will be equal to:	FA2-FA1
143.	The language {a ab aba bab} is _____ .	Regular
144.	Decomposing a string into its valid units is referred as:	Tokenizing
145.	Let FA3 be an FA corresponding to FA1+FA2, then the initial state of FA3 Must corresponds to the initial state of	FA1 or FA2
146.	If FA1 corresponds to (a+b)* then FA1 must accept _____ string/strings. Select correct option: No Odd length	EVERY
147.	A regular language can be:	irregular infinite non-deterministic **None of the given options**
148.	N into its valid units is referred as:	Tokenizing
149.	The strings of FA2 are accepted first before the strings of FA1	Palindromes
150.	There _____ a language for which only FA can be built but not the RE.	Be may
151.	Kleene's Theorem part I expresses the relationship between ___	FA and TG

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CS-402 Important Mcq's For Mid Term !! Solve By Vu-Topper RM!!



Question No:1 **(Marks:1)** **Vu-Topper RM**

Σ={a,Aa,Abb}, then string aAaAbbAa has__length.

A. One
B. Two
C. Three
D. Four **Page 4**

Question No:2 **(Marks:1)** **Vu-Topper RM**

Languages generated by kleene star are always___.

A. Finite
B. Infinite **Page 7**
C. Sometimes finite & sometimes infinite
D. None of the these

Question No:3 **(Marks:1)** **Vu-Topper RM**

Let S = {aa, bb}, then S* will have the _____ string.

A. Λ **Page 7**
B. abba
C. aabbbaa
D. bbaab

Question No:4 **(Marks:1)** **Vu-Topper RM**

If r1 = (aa + bb) and r2 = (a + b) then the language (aa + bb)* will be generated by

A. (r1)(r2)
B. (r1 + r2)
C. (r2)*
D. (r1)* **Page 10**

Question No:5 **(Marks:1)** **Vu-Topper RM**

If a language can be expressed through FA, then it can also be expressed through TG.

A. True **Page 25**
B. False

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Question No:6 **(Marks:1)** **Vu-Topper RM**

If an alphabet has n number of letter, then number of strings of length m will be

A. n+m
B. (n)(m)
C. m^n
D. n^m Page 6

Question No:7 **(Marks:1)** **Vu-Topper RM**

In GTG, if a state has more than one incoming transitions from a state. Then all those incoming transitions can be reduced to one transition using sign

A. -
B. + Page 27
C. *
D. ( )

Question No:8 **(Marks:1)** **Vu-Topper RM**

Above given FA accepts___strings defined over Σ={a , b}

A. All Page 15
B. Some
C. All but not null
D. None of these

Question No:9 **(Marks:1)** **Vu-Topper RM**

One FA has 3 states and 2 letters in the alphabet. Then FA will have _____number of transitions in the diagram

A. 4
B. 5
C. 7
D. 6 Page 14

Question No:10 **(Marks:1)** **Vu-Topper RM**

Every FA should be

A. Deterministic Page 25
B. Non- Deterministic
C. Deterministic & Non- Deterministic
D. None of these

Question No:11 **(Marks:1)** **Vu-Topper RM**

Auto Meta mean

A. Manual work

B. Automatic work **Page 3**

Question No:12 **(Marks:1)** **Vu-Topper RM**

NFA to FA will

A. Equal **Page 43**

B. Not equal

C. Not valid

D. None of given

Question No:13 **(Marks:1)** **Vu-Topper RM**

The length of output string in case of__is one more than the length of corresponding input string.

A. Finite Automaton **Page 55**

B. TG

C. GTG

Question No:14 **(Marks:1)** **Vu-Topper RM**

The_machine helps in building a machine that can perform the addition of binary numbers.

A. Incrementing **Page 60**

B. Complementing

C. Decrementing

D. None of the given

Question No:15 **(Marks:1)** **Vu-Topper RM**

There _____ be dead states in NFA.

A. may not

B. must

C. should not

D. **will**

Question No:16 **(Marks:1)** **Vu-Topper RM**

1 Let FA1 accepts many strings and FA2 accepts none then FA1+FA2 will be equal to:

A. FA1

B. FA2
C. FA2-FA1
D. (FA2)

Question No:17 **(Marks:1)** **Vu-Topper RM**
If FA1 corresponds to (a+b)* then FA1 must accept _____ string/strings.
A. No
B. Odd length
C. Even length
D. **Every**

Question No:18 **(Marks:1)** **Vu-Topper RM**
A regular language can be:
A. irregular
B. infinite
C. non-deterministic
D. **None of the given options**

Question No:19 **(Marks:1)** **Vu-Topper RM**
There _____ a language for which only FA can be built but not the RE.
A. is cannot be
B. may be
C. may not be

Question No:20 **(Marks:1)** **Vu-Topper RM**
For every three regular expressions R, S, and T, the languages denoted by R(S U T) and (RS) U (RT) are the _____ .
A. **Same**
B. Different

Question No:21 **(Marks:1)** **Vu-Topper RM**
In _____ there must be transition for all the lettersof a string.
A. NFA
B. GTG
C. TG
D. **FA**

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Question No:22 **(Marks:1)** **Vu-Topper RM**

We cannot construct an NFA for the language of _____ defined over alphabet set {a,b}.

A. Even
B. odd
C. **Palindromes**
D. Integers

Question No:23 **(Marks:1)** **Vu-Topper RM**

Decomposing a string into its valid units is referred as:

A. Decomposing
B. Splitting
C. **Tokenizing**
D. Dividing

Question No:24 **(Marks:1)** **Vu-Topper RM**

In concatenation, we include the initial state of FA2 automatically after the final state of FA1 because of:

We need just one initial state

Question No:25 **(Marks:1)** **Vu-Topper RM**

Considering FA1 and FA2 having 2 states each. Now FA1+FA2 can have maximum _____ number of states.

A. 2
B. 3
C. **more than 3**
D. None of these

Question No:26 **(Marks:1)** **Vu-Topper RM**

If R is a regular language and L is some language, and L U R is a _____, then L must be a _____.

A. **Regular language**
B. Finite Auto

Question No:27 **(Marks:1)** **Vu-Topper RM**

The minimum length of the strings(except null string) of a language that starts and ends in different letters will be:

A. **1**

B. 2
C. 3
D. 4

Question No:28 **(Marks:1)** **Vu-Topper RM**

Consider we have languages L7 and L6. Which of the following represents their concatenation?

A. **L7+L6**
B. L7/L6
C. L6L7
D. L7L6

Question No:29 **(Marks:1)** **Vu-Topper RM**

Let FA1 has x number of states and FA2 has y number of states. Now FA1+FA2 can have maximum _____ **number of states.**

A. x+y
B. x-y
C. x/y
D. **none**

Question No:30 **(Marks:1)** **Vu-Topper RM**

The language {a ab aba bab} is _____ .

A. Irregular
B. **Regular**
C. Recursive
D. None of the given options

Question No:31 **(Marks:1)** **Vu-Topper RM**

If we have a finite language and the number of states in the FA is n then the maximum number of letters in the each word of the language that will be accepted by the given FA will be:

A. N
B. n-1
C. n+1
D. 1

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Question No:32 **(Marks:1)** **Vu-Topper RM**

Moore machine can have -------- final states.

A. 2
B. 4
C. 6
D. 8

Question No:33 **(Marks:1)** **Vu-Topper RM**

Let L be the language of all strings, defined over Σ = {0,1}, ending in 111. Which of the following strings are distinguishable with respect to L with z being 11?

111, 101

Question No:34 **(Marks:1)** **Vu-Topper RM**

Let L be the language of all strings, defined over ∑ = {0,1}, ending in 10. Which of the following strings are distinguishable with respect to L with z being 0?

A. 010, 101
B. 111, 101
C. 001, 101
D. 111, 111

Question No:35 **(Marks:1)** **Vu-Topper RM**

There _____ be a unique path for each valid string (called a word) in NFA.

A. May not
B. Must
C. Should not
D. Will

Question No:36 **(Marks:1)** **Vu-Topper RM**

If we have only one state, having no transition for input letters, then it is an example of:

A. RE
B. FA
C. TG
D. NFA

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Question No:37 **(Marks:1)** **Vu-Topper RM**

Which of the following state is introduced while developing NFA for the closure of an FA?

An initial state which should be final as well

Question No:38 **(Marks:1)** **Vu-Topper RM**

A _____ with "n" states must accept at least one string of length greater than "n".

A. DFA
B. RE
C. Irregular language
D. Irrelevant language

Question No:39 **(Marks:1)** **Vu-Topper RM**

In Moore machine, output is produced over the change of:

A. Transitions
B. Transitions and states
C. None of the mentioned
D. States

Question No:40 **(Marks:1)** **Vu-Topper RM**

In NFA,if null word (lambda) is allowed to be a label of an edge, then that NFA is called _____.

NFA with null string

Question No:41 **(Marks:1)** **Vu-Topper RM**

Every _____ can be considered to be _____ as well, but the converse may not be true.

A. TG, FA Page 19
B. GTG
C. PDA
D. FA, TG

Question No:42 **(Marks:1)** **Vu-Topper RM**

In the context of make NFA for the concatenation of FA1 and FA2 (Both FAs accepting null string), which of the following option is correct?

A. Final states in both FAs
B. Initial states in both FAs

C.
D. FA2 having initial state only
E. FA2 having final state only

Question No:43 **(Marks:1)** **Vu-Topper RM**

In order to make NFA for the union of FA1 and FA2, the new initial state should be linked to:

A. Initial states of both FAs
B. Initial and final states of FA1 and FA2 respectively
C. Initial state of FA1 only
D. Final and initial states of FA1 and FA2 respectively

Question No:44 **(Marks:1)** **Vu-Topper RM**

Keeping in view the discussion by Martin, how many states are required to recognize the language of all strings of length 2 or more defined over ∑= {a,b}, with 'b' being the second letter from right?

A. 9
B. 6
C. 7
D. 8

Question No:45 **(Marks:1)** **Vu-Topper RM**

Let FA3 be an FA corresponding to FA1FA2, then the final state of FA3 must correspond to the final state of

FA2 only

Question No:46 **(Marks:1)** **Vu-Topper RM**

Let FA3 be an FA corresponding to FA1FA2, then initial state of FA3 must correspond to the initial state of

A. FA1 only
B. FA2 only
C. FA1 and FA2
D. FA1 or FA2

Question No:47 **(Marks:1)** **Vu-Topper RM**

In which of the following machine, the length of output string is the same to that of input string?

A. Mealy machine

B. Moore machine
C. Finite automaton with output
D. Non-deterministic finite automaton

Question No:48 **(Marks:1)** **Vu-Topper RM**
Moore Machine is an application of:
A. Finite automata with output
B. Finite automata with input
C. None

Question No:49 **(Marks:1)** **Vu-Topper RM**
---------- state is not important in Moore machine.
Final

Question No:50 **(Marks:1)** **Vu-Topper RM**
In Mealy machine the output depends on _____
A. Present state and Present input
B. Only present state
C. Nothing
D. Type of input

Question No:51 **(Marks:1)** **Vu-Topper RM**
If L is a regular language, then (L')' U L will be:
A. L
B. C
C. P
D. F

Question No:52 **(Marks:1)** **Vu-Topper RM**
A string will be accepted by an NFA if there exists _____ one successful path.
A. Atleast
B. Atmost
C. Maximum
D. None of the given options

Question No:53 **(Marks:1)** **Vu-Topper RM**
If A and B are regular languages, !(A' U B') is:

A. Non regular
B. May be regular
C. None of the mentioned
D. Regular

Question No:54 **(Marks:1)** **Vu-Topper RM**

There is no question of accepting any language in:

A. Moore machine
B. FA
C. TG
D. GTG

Question No:55 **(Marks:1)** **Vu-Topper RM**

Let FA3 be an FA corresponding to FA1FA2, then the initial state of FA3 must correspond to the initial state of

FA1 or FA2

Question No:56 **(Marks:1)** **Vu-Topper RM**

Let FA3 be an FA corresponding to FA1FA2, then final state of FA3 must correspond to the final state of

A. FA2 only
B. FA1 only
C. FA1 or FA2
D. FA1 and FA2

Question No:57 **(Marks:1)** **Vu-Topper RM**

If we subtract a binary number 1010 from the binary number 1101(ignore the overflow), then the result will be:

1100

Question No:58 **(Marks:1)** **Vu-Topper RM**

Subtraction of binary numbers is possible through:

A. Both complementing and incrementing machine
B. Complementing machine
C. Incrementing machine
D. Converting machine

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Question No:59 **(Marks:1)** **Vu-Topper RM**

For a given Moore Machine, the input string is '101010', thus the output string would be of length:

A. Length of input string + 1
B. Length of input string – 1
C. Length of input string + 2
D. Length of input string -2

Question No:60 **(Marks:1)** **Vu-Topper RM**

Which one of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?

A. Mealy machine
B. Moore machine
C. Finite state machine
D. Deterministic finite state machine

Question No:61 **(Marks:1)** **Vu-Topper RM**

If L1 and L2′ are regular languages, L1 ∩ (L2′ U L1′)’ will be

A. Regular **Page 10**
B. Ir-regular
C. Can’t be decided
D. Another Language which is not listed here

Question No:62 **(Marks:1)** **Vu-Topper RM**

Mealy machine is equivalent to Moore machine, if we:

Applications of complementing and incrementing machines

Question No:63 **(Marks:1)** **Vu-Topper RM**

Given the language L = {ab, aa, baa}, which of the following strings are inL*?
abaabaaabaa
aaaabaaaa
baaaaabaaaab
baaaaabaa

A. 1, 2 and 3
B. 2, 3 and 4
C. 1, 2 and 4

D. 1, 3 and 4

Question No:64 **(Marks:1)** **Vu-Topper RM**

In the context of make NFA for the concatenation of FA1 and FA2 (FA2 accepting null string), which of the following option is correct?

Final states in both FAs

Question No:65 **(Marks:1)** **Vu-Topper RM**

In the context of make NFA for the concatenation of FA1 and FA2 (none accepting null string), which of the following option is correct?

No initial state in FA1 only

Question No:66 **(Marks:1)** **Vu-Topper RM**

Two machines are said to be equivalent if they print the_outputstring when the _____input string is run on them.

A. Same, Same
B. Same, different
C. Different, same
D. Unique, different

Question No:67 **(Marks:1)** **Vu-Topper RM**

Every NFA can be considered to be a - as well, but the converse may not be true.

A. TG
B. FA
C. GTG
D. PDA

Question No:68 **(Marks:1)** **Vu-Topper RM**

In which of the following machine, the length of output string is 1 more than that of input string?

A. Mealy machine
B. Non-deterministic finite automaton
C. Finite automaton with output
D. Moore machine

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Question No:69 **(Marks:1)** **Vu-Topper RM**

If S = {aa, bb} then S* will not contain _____.

A. abbbab

B. bbba

C. bbbbab

D. ababbb

Question No:70 **(Marks:1)** **Vu-Topper RM**

Which of the following machine has only one initial state and no finalstate?

A. Moore machine

B. Finite state machine

C. Deterministic finite state machine

Question No:71 **(Marks:1)** **Vu-Topper RM**

Which of the following diagram is very rigid in order to express any language?

A. TG

B. NFA

C. GTG

D. FA

Question No:72 **(Marks:1)** **Vu-Topper RM**

If S = {a}, then S+ will be

A. {a, aaa, aaaa, aaaaa,…}

B. {a, aa, aaa, aaaa,…}

C. {a, aaa, aaaaa, aaaaaaa,…}

D. {aa, aaaa, aaaaaa, aaaaaaaa,…}

Question No:73 **(Marks:1)** **Vu-Topper RM**

Let L be the language of all strings. defined over ∑ = {0,1}. ending in 111. Melay machine can have final states.

A. Zero

B. One

C. More than one but finite

D. More than one but infinite

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Question No:74 **(Marks:1)** **Vu-Topper RM**

Let's we have two regular expressions R1=(xx+yy) and R2=(x+ y). Whichone of the following is the correct regular expression for the Union of R1 and R2?

A. (xx+yy)(x+y)
B. (xx+yy)+(x+y)*
C. (xx+yy)+(x+y)
D. ((xx+yy)+(x+y))*

Question No:75 **(Marks:1)** **Vu-Topper RM**

The state where there is no way to leave after entry, is called _____.

A. Davey John locker
B. initial state
C. final state
D. non-final state

Question No:76 **(Marks:1)** **Vu-Topper RM**

Which one of the following word is not accepted by the given regular expression?

A. aaabab
B. aaaababb
C. abbaab
D. aabbabb

Question No:77 **(Marks:1)** **Vu-Topper RM**

According to theory of automata there are___types of languages

A. One
B. Two
C. Three
D. Four

Question No:78 **(Marks:1)** **Vu-Topper RM**

Regular languages are closed under the following operations.

A. Union only
B. Concatenation, Closure only
C. Union, Concatenation and Closure

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Question No:79 **(Marks:1)** **Vu-Topper RM**

Regular languages are closed under the following operations.

A. Union only
B. Concatenation, Closure only
C. Union, Concatenation and Closure
D. Regular languages are not closed under any operation

Question No:80 **(Marks:1)** **Vu-Topper RM**

There can be more than____FA for a certain language but for_FAthere is only one language associated with it:

A. one, one
B. one, two
C. two, three
D. two, one

Question No:81 **(Marks:1)** **Vu-Topper RM**

There is one compulsion that each state must have an on outgoing edge for every input variable in:

A. Finite Automata
B. Transition Graph
C. Both Finite Automata and Transition Graph
D. Transition Table

Question No:82 **(Marks:1)** **Vu-Topper RM**

FA is also called

A. TG
B. GTG
C. NFA
D. DFA

Question No:83 **(Marks:1)** **Vu-Topper RM**

If r1 and r2 are regular expressions then (r1 * r2) is _____ .

A. FA
B. TG
C. GTG
D. RE

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Question No:84 **(Marks:1)** **Vu-Topper RM**

Keep in view the language of all strings ending with 'a' defined over Σ = {a, b, c, d}. For which input letter, we will take a loop on the final state of its transition diagram?

A. a
B. b
C. c
D. d

Question No:85 **(Marks:1)** **Vu-Topper RM**

Which of the following statements is true about NFA with Null String?

A. Infinite states
B. Infinite set of letters
C. Infinite set of transitions
D. Transition of null string is allowed at any stage

Question No:86 **(Marks:1)** **Vu-Topper RM**

Introducing new start state in case of multiple start states is the step no. of proving Kleene's theorem part ||.

A. 1
B. 2
C. 3
D. 4

Question No:87 **(Marks:1)** **Vu-Topper RM**

Which of the following diagrams expresses languages more simply?

A. FA
B. NFA
C. TG
D. GTG

Question No:88 **(Marks:1)** **Vu-Topper RM**

The language of all strings defined over alphabet set = {a, b} that does notend with 'a' actually ends with:

A. b
B. b and ^
C. ^

D. ^ and a

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Question No:89 **(Marks:1)** **Vu-Topper RM**

In NFA having no transition at certain state, FA can be built byintroducing:

A. Empty state
B. Combination of states
C. Initial state
D. Final state

Question No:90 **(Marks:1)** **Vu-Topper RM**

Formal is also known as

A. Syntactic language
B. Semantic language
C. Informal language
D. None of these

Question No:91 **(Marks:1)** **Vu-Topper RM**

There may be more than one transition for a certain letter on a state in:

A. Finite automata
B. Non-Deterministic Finite Automata
C. Transition Table
D. Moore Machine

Question No:92 **(Marks:1)** **Vu-Topper RM**

FA of EVEN language shows null string when

A. Initial state is final as well
B. EVEN does not accept null
C. One state is declared null
D. None of the these

Question No:93 **(Marks:1)** **Vu-Topper RM**

Which of the following statement is true about GTG?

A. Transitions are based on input letters
B. Transitions are based on specified substrings
C. Transitions are based on regular expressions
D. Transitions are based on alphabet set

Question No:94 **(Marks:1)** **Vu-Topper RM**

In GTG, there can be more than one:

A. Start state
B. Final state
C. Start state and final state
D. Null state

Question No:95 **(Marks:1)** **Vu-Topper RM**

GTG for the expression (aa+aba)* may have minimum number of states:

A. 1
B. 2
C. 3
D. 4

Question No:96 **(Marks:1)** **Vu-Topper RM**

In regular expressions, the operator '*' stands for

A. Concatenation
B. Iteration
C. Selection
D. Add

Question No:97 **(Marks:1)** **Vu-Topper RM**

If r1 is a regular expression then (r1)* is____.

A. A generalized transition graph
B. A non-deterministic finite automaton
C. A finite automaton
D. Also, a regular expression

Question No:98 **(Marks:1)** **Vu-Topper RM**

Which of the following is the bypass and state elimination step in thecontext of Kleene's theorem part || proof?

A. 1
B. 2
C. 3
D. 4

Question No:99 **(Marks:1)** **Vu-Topper RM**

Which of the following is free of non-determinism?

A. TG

B. FA
C. NFA
D. NFA-^

Question No:100 **(Marks:1)** **Vu-Topper RM**
Melay machine to increase the output string in magnitude by 1 is called:
A. Complementing machine
B. Incrementing machine
C. Decrementing machine
D. Converting machine

Question No:101 **(Marks:1)** **Vu-Topper RM**
A. Kleene's Theorem Part I expresses the relationship between_____.
B. FA and TG
C. TG and RE
D. RE and FA
E. FA and RE

Question No:102 **(Marks:1)** **Vu-Topper RM**
Suppose we have FA3 (which is equal to FA1 + FA2), then the final stateof FA3 will be declared final if:
A. It corresponds to final states of both FA1 and FA2
B. It corresponds to final states of FA1 only
C. It corresponds to final states of FA2 only
D. It corresponds to any of the final states in FA1 or FA2

Question No:103 **(Marks:1)** **Vu-Topper RM**
Null strings can be specified on edges in:
A. Finite Automata
B. Non-Deterministic Finite Automata
C. Transition Graph
D. Melay Machine

Question No:104 **(Marks:1)** **Vu-Topper RM**
What is false about the PALINDROME LANGUAGE?
A. Every word is reverse of itself.
B. It is an infinite language.
C. FA can be build for it.

D. None of the given option

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Question No:105 (Marks:1) Vu-Topper RM

While finding RE corresponding to TG, If TG has more than one startstate then

A. Introduce the new start state

B. Eliminate the old start state

C. Replace the old start stat with final state

D. Replace the old final state with new start state

Question No:106 (Marks:1) Vu-Topper RM

All possible combinations of strings of a language including null string is referred as:

A. Concatenation of a language with itself

B. Kleene star closure of a language

C. Multiplication of language with itself

D. Addition of a language with itself

Question No:107 (Marks:1) Vu-Topper RM

n! will be equal to:

A. n*n

B. n*(-n)!

C. n*(n-1)

D. n*(n-1)!

Question No:108 (Marks:1) Vu-Topper RM

While finding RE corresponding to a TG, we connect the new start statewith the old start state by__transition.

A. a

B. b

C. Null

D. RE

Question No:109 (Marks:1) Vu-Topper RM

In proving Kleene Theorem II, if three states are connected then middle state is removed by connecting first and third state and writing corresponding RE in:

A. Sum

B. Concatenation

C. Difference
D. Asterisk

Question No:110 **(Marks:1)** **Vu-Topper RM**

In___there must be transition for all the letters of a string.

A. NFA
B. GTG
C. TG
D. FA

Question No:111 **(Marks:1)** **Vu-Topper RM**

There is no question accepting any language in:

A. FA
B. TG
C. GTG
D. Moore machine

Question No:112 **(Marks:1)** **Vu-Topper RM**

The FA can be drawn for the regular expression (a+b)* with minimum state(s).

A. 1
B. 2
C. 3
D. 4

Question No:113 **(Marks:1)** **Vu-Topper RM**

Which of the following does not contribute while finding out the length of strings?

A. ^
B. a
C. b
D. a+b

Question No:114 **(Marks:1)** **Vu-Topper RM**

The language of all strings defined over alphabet set = {x, y} that endswith same letters will have the maximum length of:

A. 1
B. 2

C. 3

D. Infinite

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Question No:115 **(Marks:1)** **Vu-Topper RM**

Considering FA1 and FA2 states each. Now FA1+FA2 can have maximum _____number of states.

A. 2

B. 3

C. More than 3

D. None of the given option

Question No:116 **(Marks:1)** **Vu-Topper RM**

Which one of the following is the RE for the language defined over ∑= {a,b} having all the words starting with a?

A. (a + b)*

B. aa(a + b)+

C. a(a + b)*

D. a*(a + b)

Question No:117 **(Marks:1)** **Vu-Topper RM**

An__can be considered to be an intermediate structure between Finite automaton and Transition Graph.

A. RE

B. GTG

C. NFA

D. None of the given options

Question No:118 **(Marks:1)** **Vu-Topper RM**

Suppose a language L1 has 2 states and L2 has2 states. If we have a machine M that accepts L1 ∩ L2. Then, the total number of states in M isequal to

A. 2

B. 4

C. 6

D. 8

Question No:119 **(Marks:1)** **Vu-Topper RM**

FA corresponding to an NFA can be built by introducing an empty state fora letter having

A. No transition at certain state
B. One transition at certain state
C. Two transitions at certain state
D. More than two transitions at certain state

Question No:120 **(Marks:1)** **Vu-Topper RM**
Automata is the plural of___.
A. Automate
B. Automaton
C. Automation
D. Automatic

Question No:121 **(Marks:1)** **Vu-Topper RM**
In NFA having no transition at certain. FA can be built by introducing:
A. Empty state
B. Combination of states
C. Initial state
D. Final state

Question No:122 **(Marks:1)** **Vu-Topper RM**
If S = { x }, then S* will be _____.
A. {^,x,xxx,xxxx,xxxxx,…}
B. {^,x,xx,xxx,xxxx,…}
C. {^,x,xxx,xxxxx,xxxxxxx,…}
D. {^,xx,xxxx,xxxxxx,xxxxxxxx,…}

Question No:123 **(Marks:1)** **Vu-Topper RM**
In TG, the string is supposed to be _____ if there is no path for a string from initial to final state.
A. Accept null string
B. Accept all strings
C. Accept all non-empty strings
D. Does not accept any string

Question No:124 **(Marks:1)** **Vu-Topper RM**
In Moore machine, if the length of input string is 9, then the length of output string will be:
A. 7

B. 8
C. 9
D. 10

Question No:125 **(Marks:1)** **Vu-Topper RM**

When ODD language is expressed by an FA, then it will have minimum states.

A. One
B. Two
C. Three
D. Four

Question No:126 **(Marks:1)** **Vu-Topper RM**

[(a + b)(a + b)]*, given RE cannot generate the string __.

A. abbaabab
B. abbbaa
C. bbbbbb
D. abbbaaaaa

Question No:127 **(Marks:1)** **Vu-Topper RM**

The recursive method for defining a language has_steps.

A. One
B. Two
C. Three
D. Four

Question No:128 **(Marks:1)** **Vu-Topper RM**

Consider the following RE:
a(a + b)b*
All of the following words are accepted except___.

A. aab
B. abb
C. aa
D. aba

Question No:129 **(Marks:1)** **Vu-Topper RM**

For every three regular expressions R, S, T, the languages denoted byR(S ∪ T) and (RS) ∪ (RT) are the___.

A. Same

B. Different
C. R(S ∪ T) is greater
D. None of the given options

Question No: 130 (Marks:1) Vu-Topper RM

Alphabet S = {a, bc, cc} has_____number of letters.

One
Two
Three
Four

Question No:131 (Marks:1) Vu-Topper RM

Two FAs are said to be equivalent, if they

A. Accept null string
B. Accept same language
C. Accept different language
D. None of the given options

Question No:132 (Marks:1) Vu-Topper RM

_____can also help in proving Kleene Theorem III.

A. NFA
B. PDA
C. Moore machine
D. Melay machine

Question No:133 (Marks:1) Vu-Topper RM

Kleene's Theorem Part II expresses the relationship between _____.

A. FA and TG
B. TG and RE
C. RE and FA
D. FA and RE

Question No:134 (Marks:1) Vu-Topper RM

If two RE's generate same language then these RE's are called____.

A. Same RE
B. Equal RE
C. Similar RE
D. Equivalent RE

Question No:135 **(Marks:1)** **Vu-Topper RM**

Every FA should be__.

A. Deterministic

B. Non-deterministic

C. Deterministic and non-deterministic

D. Not depends on language

Question No:136 **(Marks:1)** **Vu-Topper RM**

What statement is true?

A. A letter is always a combination of symbols

B. A letter may consist of one symbol

C. There is no difference between symbol and letter

D. Letters and symbols are the same thing

Question No:137 **(Marks:1)** **Vu-Topper RM**

If ∑= {ab, bb}, then ∑* will not contain

A. abbbab

B. bbba

C. bbbbab

D. ababbb

Question No:138 **(Marks:1)** **Vu-Topper RM**

Choose the correct word produced by RE (a + b)* ab

A. abb

B. abab

C. bbbb

D. aaaa

Question No:139 **(Marks:1)** **Vu-Topper RM**

According to 1st part of the Kleene's theorem, If a language can beaccepted by an FA then it can be accepted by a_as well

A. FA

B. CFG

C. GTG

D. TG

Question No:140 **(Marks:1)** **Vu-Topper RM**

"One language can be expressed by___GTG".

A. Only one
B. Only two
C. More than one

Question No:141 **(Marks:1)** **Vu-Topper RM**

If a TG has more than one start states, then we can make a single startstate by introducing a new state and connecting it with all the previously existing start states by using.

A. Any infinite string
B. Single letter string
C. Null string
D. Any finite string

Question No:142 **(Marks:1)** **Vu-Topper RM**

If in a NFA, ^ is allowed to be a label of an edge then that NFA iscalled _____.

A. TG
B. RE
C. NFA with null string
D. RE

Question No:143 **(Marks:1)** **Vu-Topper RM**

If we want to make a Moore machine equivalent to mealy machinethen

A. We should ignore the extra character printed by the Moore machine.
B. We should ignore the extra character printed by the Mealy machine.
C. We will make the initial state as a no carry state.
D. We should not ignore the extra character printed by the Moore machine.

Question No:144 **(Marks:1)** **Vu-Topper RM**

Two machines are said to be equivalent if they print the output string when same input string is run no them.

A. Same
B. Different
C. Inverse
D. Null

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Question No:145 **(Marks:1)** **Vu-Topper RM**

The length of output in case of___is one more than the length of corresponding input string

A. Moore machine
B. Mealy machine
C. Incremental machine
D. Adding machine

Question No:146 **(Marks:1)** **Vu-Topper RM**

A is not a valid transition in

A. TG
B. GTG
C. NFA
D. RE

Question No:147 **(Marks:1)** **Vu-Topper RM**

Dead states are also called

A. John Davey Lockers
B. Davey John Lockers
C. Mutex Lockers
D. Semaphores

Question No:148 **(Marks:1)** **Vu-Topper RM**

Language of all strings whose length is odd and number of y's even defined over alphabet set ∑ = {x, y} .__will be accepted by the given language.

A. xxyxyxyyyx
B. xxyxyxyyyxy
C. xxyxyxyyyxx
D. xxyxyxyyy

Question No:149 **(Marks:1)** **Vu-Topper RM**

If an effectively solvable problem has answer in Yes or NO. then the solution is called

A. Infinite problem

B. Decision procedure
C. Finite solution
D. Optimal procedure

Question No:150 **(Marks:1)** **Vu-Topper RM**
If the intersection of two regular languages is regular then thecomplement of the intersection of these two languages is
A. Regular
B. Irregular
C. Irregular but finite
D. Irregular but infinite

Question No:151 **(Marks:1)** **Vu-Topper RM**
If R is regular language and Q is any language (regular/non-regular). Then Pref(_in___) is regular.
A. Q, Q
B. Q, R
C. R, Q
D. R, R

Question No:152 **(Marks:1)** **Vu-Topper RM**
The strings or words which do not belong to a language are called of that language
A. Intersection
B. Union
C. Complement
D. Quotient

Question No:153 **(Marks:1)** **Vu-Topper RM**
Prime is a_language.
A. Finite
B. Both context free and regular
C. Regular
D. Non-regular

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Question No:154 **(Marks:1)** **Vu-Topper RM**

Finite Automaton (FA) must have____number of states while alanguage has__words.

A. Infinite, finite
B. Finite, finite
C. Finite, infinite
D. Infinite, infinite

Question No:155 **(Marks:1)** **Vu-Topper RM**

The language "PRIME" is an example of___language.

A. Regular but finite
B. Regular
C. Non regular but finite
D. Non regular

Question No:156 **(Marks:1)** **Vu-Topper RM**

If L1 and L2 are regular languages then which statement is NOT true?

A. L1 + L2 is always regular
B. L1 L2 is always regular
C. L1/L2 is always regular
D. L1* is always regular

Question No:157 **(Marks:1)** **Vu-Topper RM**

If a language is regular it must generate____number of distinctclasses.

A. Finite
B. Infinite
C. Two
D. three

Question No:158 **(Marks:1)** **Vu-Topper RM**

The operators like (* . +) in the parse tree are considered as

A. Terminals
B. Non-terminals
C. Productions
D. Intermediates

Question No:159 **(Marks:1)** **Vu-Topper RM**

Set of all palindromes over {a,b} is:

A. Regular
B. Regular and finite
C. Regular and infinite
D. Non-regular

Question No:160 **(Marks:1)** **Vu-Topper RM**

Which one of the following languages is a non-regular language?

A. Even-even
B. Containing double a
C. Start and end with same letter
D. Palindrome

Question No:161 **(Marks:1)** **Vu-Topper RM**

The language of all strings partition ∑* into_class(es).

A. One
B. Two
C. Three
D. Four

Question No:162 **(Marks:1)** **Vu-Topper RM**

The language of all strings not beginning with 'b' partitions ∑* into distinct classes.

A. Two
B. Three
C. Four
D. Five

Question No:163 **(Marks:1)** **Vu-Topper RM**

The values of input (say a & b) do not remain same in one cycle dueto

A. NAND gate
B. Clock pulse
C. OR gate
D. NOT gate

Question No:164 **(Marks:1)** **Vu-Topper RM**

In a CFG, the non-terminals are denoted by

A. Small letters
B. Numbers
C. Capital letters
D. Small letters and numbers

Question No:165 **(Marks:1)** **Vu-Topper RM**

a* + b* = (a + b)* this expression is

A. True
B. False

Question No:166 **(Marks:1)** **Vu-Topper RM**

Length of EVEN-EVEN language is

A. Even
B. Odd
C. Sometimes even & sometimes odd
D. Such language doesn't exist

Question No:167 **(Marks:1)** **Vu-Topper RM**

While finding RE corresponding to TG, we connect the new start stateto the old start state by the transition labeled by

A. a
B. b
C. null
D. none of the given options

Question No:168 **(Marks:1)** **Vu-Topper RM**

Given S, Kleene star closure is denoted by

A. S*
B. S+
C. S-
D. None of these

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Question No:169 **(Marks:1)** **Vu-Topper RM**

Which of the following steps replaces multiple incoming transitionedges with a single one in proving Kleene's theorem part ||?

A. 1
B. 2
C. 3
D. 4

Question No:170 **(Marks:1)** **Vu-Topper RM**

If r1 = (aa + bb) and r2 = (a + b) then the language (aa + bb)(a + b)will be generated by

A. (r1)(r2)
B. (r1 + r2)
C. (r2)(r1)
D. (r1)*

Question No:171 **(Marks:1)** **Vu-Topper RM**

The language having even number of a's and even number of b's defined over S = {a, b} is called _____.

A. EVEN-EVEN
B. ODD-ODD
C. PALINDROME
D. FACTORIAL

Question No:172 **(Marks:1)** **Vu-Topper RM**

If L1' and L2' are regular languages. Then L1,L2 will be

A. Regular
B. Non regular
C. May be regular
D. None of the mentioned

Question No:173 **(Marks:1)** **Vu-Topper RM**

f FA1 corresponding to (a+b)* then FA1 must accept string/strings

A. No
B. Odd length
C. Even length
D. Every

Question No:174 (Marks:1) Vu-Topper RM

In FA, initial state can be represented by:

A. Drawing an arrow head before that state
B. Drawing a circle in that state
C. Drawing '+' sign in that state

Question No:175 (Marks:1) Vu-Topper RM

An FA is a collection of:

A. Finite states, finite transition and finite input letters
B. Infinite states, infinite transition and infinite input letters
C. Only finite states and finite transitions
D. Only infinite states and infinite transitions

Question No:176 (Marks:1) Vu-Topper RM

NFA with null string has initial state(s)

A. One
B. Two
C. Three

Question No:177 (Marks:1) Vu-Topper RM

The difference between number of states with regular expression (a +b) and (a + b)* is:

A. 0
B. 1
C. 2
D. 3

Question No:178 (Marks:1) Vu-Topper RM

A transition graph is converted into a(n)____in order to obtain regular expression.

A. FA
B. GTG
C. NFA
D. NFA

Question No:179 (Marks:1) Vu-Topper RM

Consider the languages L1 = and L2 = {a}. Which one of thefollowing represents L1 L2* ∪ L1*

A. ^
B. a*
C. All of the mentioned
D. None of the mentioned

Question No:180 (Marks:1) Vu-Topper RM

If S = {a, b} then which of the following RE will generate all possible strings?

A. a* + b*
B. (ab)*
C. (a + b)*
D. (ab + ba)*

Question No:181 (Marks:1) Vu-Topper RM

In drawing FA3 (which is equal to FA1 + FA2), a state will bedeclared final if

A. It corresponds to final states of both FA1 and FA2
B. It corresponds to final states of FA1
C. It corresponds to final states of FA2
D. It corresponds to any of the final states in FA1 or FA2

Let S = {a, bb, bab, baabb} be a set of strings, which one of the following will not be included in S*?

A. baba
B. baabbabb
C. bbaaabb
D. bbbaabaabb

Edges are expressed with a regular expression in:
GTG Page 23

Which one of the following word is not accepted by the given regular expression?
abbbbaa

The length of string "AbBAbcd" defined over Σ ={Ab,B,c,d} is _____.

A. One
B. Two
C. Five
D. Four

a(a+b)*b + b(a+b)*a is the regular expression of language defined over Σ={a,b} that is _____.
starting with a and ending in a

In case of finite automaton there _____ be a transition on each _____ for every letter of the alphabet set.

A. Must, state
B. May be, state
C. Often, edge
D. Must, edge

Which one of the following word is not accepted by the given regular expression?
(a+b)*(aaa+bbb)(a+b)*

A. Ababaaaab
B. Bababbbba
C. Baabaabba
D. Abbaaabba

Kleene Theorem III states that if the language can be expressed by RE then there exist -------- accepting the language.

A. FA **Page 32**
B. DFA
C. NFA
D. None

In proving Kleene Theorem II, if a state has two incoming transition edges labelled by RE from the same state, then replace all the edges with a single transition edge labelled by ------- of corresponding RE.

A. Sum **Page 27**
B. Edge
C. FA
D. RE

GTG for the expression (a+b)*bb may have minimum number of states:
Aaabcbbcbacc

Choose the correct word produced by RE (a + b)* (aa+bb).

A. Abab
B. Babab
C. aaaa
D. Ab

How many states of a finite automaton will be final for accepting the only string ‘abb’, if Σ= {a, b}?

A. 1
B. 2
C. 3
D. 4

Which one of the following is a correct word produced by the RE (a*b*)ab?
abab

FA and _____ are same except that _____ has unique symbol for each transition.

A. FA,TG

B. NFA,TG
C. NFA,FA
D. GTG,NFA

While developing NFA for the union of FA1 and FA2, if there is a loop of ‘a’ at the initial state of FA1 then the new initial state will have a transition for ‘a’ that goes straight to:
The initial state of FA1

Closure of an FA is the same as _____ of an FA with itself except that the initial state of the required FA is a final state as well.
A. Sum
B. Union
C. Intersection
D. Concatenation

How many new states are introduced while developing NFA for the closure of an FA?
A. 2
B. 4
C. 6
D. 8

In _____ there must be transitions for all the alphabets over which a language is defined.
A. FA
B. TG
C. NFA
D. GTG

Which of the following is not a step-in elimination of states procedure?
Unify single transitions to multi transitions that contains union of input

In Moore machine the output depends on
The state

Keeping in view the discussion by Martin, how many states are required to recognize the language of all strings of length 3 or more defined over ∑= {a,b}, with 'a' being the third letter from right?

A. 10
B. 12
C. 14
D. 16

Strings x,y,z belongs to Σ * such that xz ∈ L but yz ∉ L where L ⊆ Σ* are:

A. Undetermined
B. Distinguishable
C. Indistinguishable
D. Both distinguishable and indistinguishable

While developing NFA for the union of FA1 and FA2, there will be
The initial state of FA1

NFA corresponding to union of FAs is built by introducing a new start state and connect it to the states originally connected to the old start state with the --------- transitions as the old start state:
Same

If we have an NFA having 3 states, and we convert that NFA to an FA. The resultant FA will contains _____ states.

A. 1
B. 2
C. 3
D. 4

In NFA having multiple transitions at certain state, FA can be built by introducing:

A. Empty state
B. Combination of states
C. Initial state
D. Final state

CS402-THORY OF AUTOMATA

SOLVED SUBJECTIVE FROM MID TERMS PAPERS

SOLVED BY JUNAID MALIK AND TEAM



AL-Junaid Institude

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1.Which of the following steps replaces multiple incoming transition edges with a single one proving Kleene's theorem part II?

a. Step 1
b. Step 2
c. Step 3
d. Step 4

2. Let FA1 accepts many strings and FA2 accepts no strings, then FA1+FA2 will be equal to:

a.FA1

b.FA2

C.FA2-FA1

d.(FA2)*

3. Let L be the language of all strings, defined over ∑ =(0,1), ending in 10. Which of the following strings are indistinguishable w.r.t L with z being 0?

a.100.101

b.111.101

c. 110 .101

d.010.101

4. If r1=(aa+bb)and r2=(a+b) then the language (aa+bb)(a+b) will bhe generated by_____

a.(r1)(r2)

b.(r1 +r2)

c.(r2)(r1)

d.(r1)*

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5. Introducing a new start state in the case of multiple states is step no._____ of proving Kleen's theorem part II.

a.1

b.2

c.3

d.4

6. The language having even numbers of a's and even number of b's defined over S={a,b} is called _____

a. EVEN-EVEN

B.ODD-ODD

c. PALINDROME

d. FACTORIAL

7. In NFA having no transition at a certain state, FA can be built by introducing :

a.Empty state

b. Combination of states

c .initial state

d. final state

8. For every three regular expression R, S, and T. the language denoted by R(S U T) and (RS)U(RT) are the _____

a .same

b. Different

c. R(SUT)is greater

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d. none of the given options

9. Which of the following string belongs to the language of the regular expression (aa*b)?

a. baabab

b. abbbaa

c.aaaaaa

d.aabaab

10. If L1' and L2' are regular languages, then L1, L2 will be

a. non regular

b. may be regular

c. Regular

d. none of the mentioned

11. Suppose the language L1 has 2 L2 has 2 states. If we have a machine M that accepts L1∩L2. Then the total number of states in M is equal to_____

a. 2

b. 4 AL-JUNAID INSTITUTE OF GROUP

c. 6

d. 8

12. If L is a regular language, then (L')' U L will be:

a. L

b.

c.

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d.

13. In Mealy machine, the output depends on_____

a. Only present state

b. Present state and Present output

c. Nothing

d. Type of input

14. Strings x,y,z belongs to ∑* such that xz∈ L but yz∉ L where L ⊆∑* are:

a. indistinguishable

b. Undetermined

c. Both distinguishable and indistinguishable

d. Distinguishable

15. Melay machine to increase the output string in magnitude by 1 is called:

a. Complementing machine

b. Decrementing machine

c. Incrementing machine AL-JUNAID INSTITUTE OF GROUP

d. Converting machine

16. Suppose we have FA3 (which is to FA1+FA2) then the final state of FA3 will be declared final if:

a. it corresponds to final states of both FA1 and FA2

b. it corresponds to final states FA1 only

c. it corresponds to final states FA2 only

d. It corresponds to any of the final states of FA1 or FA2

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17. If we have the finite language and the number of states in the FA is n then the maximum number of letter in each word of the language that will be

accepted by the given FA will be :

a. 1

b. n+1

c. n-1

d. n

18. Which of the following state is introduced while developing NFA for the closure of FA?

a. An initial state which should be final as well

b. simply an initial AL-JUNAID INSTITUTE OF GROUP

c. Final states

d. An initial state with loop for all letters

19. Length of EVEN-EVEN language is_____

a. Odd

b. such language doesn't exist

c. Even

d. sometimes even & sometimes odd

20. If FA1 corresponds to (a+b)* then FA1 must accept_____ strings/strings.

a. No

b. Odd length

c. Every

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d. Even length

21. In FA, initial stage is represented by :

a. leaventh state empty

b. drawing a circle in that state

c. drawing '+'sign in transtate

d. Drawing an arrowhead before the state

22. Which one of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?

a. Moore machine

b. Finite state machine AL-JUNAID INSTITUTE OF GROUP

c. Mealy machine

d. Deterministic finite state machine

23. Length of machine "AbBAbcd" defined over ∑=(Ab,B,c,d) is_____.

a. three

b. four

c. five

d. six

24. An FA is a collection of:

a. Finite states , finite transition and finite input letters

b. infinite states , infinite transition and infinite input letters

c. Only finite states , finite transition

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d. Only infinite states , infinite transition

25. Given the language L={ab, aa, baa}, which of the following strings are in L*?

i) abaabaaabaa
ii) aaaabaaaa
iii)baaaaabaaaab
iv)baaaaabaa

a. 1,2 & 3

b. 2, 3&4

c. 1,2 & 4 AL-JUNAID INSTITUTE OF GROUP

d. 1, 3&4

26. In the context to make NFA for the concentration of FA1 and FA2 (FA1 accepting null string) which of the following option is correct?

a. initial states in both FAs

b. FA2 having initial state only

c. FA2 having final state only

d. Final states in both FAs

27. Every ___ is a ___ as well, but the converse may not be true.

a. TG ,FA

b. TG, GTG

c. FA ,GTG

d. FA, TG

28. NFA with null string has ____ initial state(s).

a. One

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b. Two

c. Three

d. Many

29. While finding RE corresponding to a TG, we connect the new start state with the old start state by____ transition.

a. a

b. b

c. Null

d. Re

30. If S=(x). then S* will be_____.

a. {^,x,xxx,xxxx,xxxx….}

b. {^,x,xxx,xxxx,xxxxx….}

c. {^,x,xx,xxx,xxxx….}

d. {^,x,xxxx,xxxxx,xxxxxx….}

31. The minimum length of string (except null string) of a language that starts and ends in the same letter will be:

a.1 AL-JUNAID INSTITUTE OF GROUP

b. 2

c.3

d. 4

32. If S={ab, bb} , then S* will not contain_____.

a. abbbab

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b. bbbbab

c. ababbb

d. bbba

33. Which of the following machine has only one initial state?

a. Moore machine

b. Finite state machine

c. Deterministic finite state machine

d. Non deterministic finite state machine

34. Which of the following diagram is very rigid in order to express any language?

a. TG

b NFA

c GTG

d FA

35. Let L be the language of all strings, defined over ∑ =(0,1), ending in 111. Which of the following strings are indistinguishable w.r.t L with z being 11?

a 111, 101

b 100, 101

c110, 101

d 010, 101

36. Mealy machine can have _____ final states.

a. One

b More than one but finite

c More than one but infinite

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d. Zero

37. In Moore machine , output is produced over the change of:

a. transition

b. transition and states

c. States

d. None of the mentioned

38. Lets we have two regular expressions R1 ={xx+yy}and R2={x+y} . Which one of the following is the correct regular expression for the union of R1 and R2?

a. (xx +yy) (x +y)

b. (xx +yy) +(x +y)*

c. (xx+ yy) +(x+ y)

d. ((xx +yy)+(x+ y))*

39. FA corresponding to an NFA can be built by introducing a state corresponding to the combination of states, for a letter having

a. no transition at certain state AL-JUNAID INSTITUTE OF GROUP

b. one transition at certain state

c. more than one transitions at certain state

d. none of the given options

40. The situation there is no way to leave after entry is called

a Davey John locker

b. initial state

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c. finial state

d. non- final state

41. Which one of the following word is not accepted by the given regular expression?

a. aaabab

b. aaaababb

c. abbaab

d. aabbabb

42. According to the theory of automata, there are _____ types of languages.

a. one

b. Two

c. three

d. four

43. Which of the following word is not accepted by the given regular expression?

a. ababaaaab

b. bababbbba

c. baabaabba

d. abbaaabba

44. Regular languages are closed under the following operations.

a. Union only

b. Concatenation, closure only

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c. Union, Concatenation and closure

d. Regular languages are not closed under any operation.

45. There can be more than ____ FA for a certain language but for ____FA there is only one language associated with it.

a. one, one AL-JUNAID INSTITUTE OF GROUP

b. one, two

c. two, three

d. two ,one

46. There is no compulsion that each state must have an on outdoing edge for every input variable in:

a. Finite automata

b. Transition graph

c. Both finite Automata and Transition Graph

d. Transition Table

47. FA is also called_____.

a. TG

b. GTG

c. NFA

d. DFA

48. If r1 and r2 are regular expressions then (r1*r2) is_____

a. FA

b. RE (Regular expression

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c. TG

d. GTG

49. Which of the following is the minimal number of states for a finite automaton accepting the language of all strings defined over any alphabet set?

a. 1

b. 2

c. 3

d. 4

50. Keeping view language of all strings ending with 'a' for which symbol we will take a loop on the final state of its transition diagram?

a. A AL-JUNAID INSTITUTE OF GROUP

b. B

c. C

d. D

51. Which of the following statements is true about NFA with Null String?

a. Infinite state

b. Infinite set of letters

c. Infinite set of transitions

d. Transition of null string is allowed at any stage

52. Which one of the following diagrams expresses languages more simply?

a. FA

b. NFA

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c. TG

d. GTG

53. The language of all strings defined over alphabet set={a,b} that does not end with 'a' actually end with:

a. a and ^

b. b

c. b and^

d ^

54. Let S={aa,bb}, then S* will have the _____ string.

a. abba

b. aabbbaa

c. bbaab

d. ^

55. Formal is also known as_____.

a. Syntactic language

b. Semantic language

c. informal language

d. none of these

56. There may be more than one transition for a certain letter on a state in:

a. Non-Deterministic finite automata

b. finite automata

c. transition table

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d. Moore Machine

57. A loop at a state is supposed to be____ transition while converting Moore machine into an equivalent Maley machine.

a. incoming

b. outgoing

c. Both Incoming and incoming

d. Complex

58. FA of EVEN language shows null string when_____.

a. initial state is final as well AL-JUNAID INSTITUTE OF GROUP

b EVEN does not accept null

C One state is declared null

d. None of these

59. Which of the following statement is true about GTG?

a Transitions are based on input letters

b Transitions are based on specified substrings

c Transitions are based on regular expressions

d Transitions are based on alphabet set

60. Which of the following is the bypass and state elimination step in the context of Kleene's theorem part II proof?

a.1

b.2

c.3

d.4

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61. Which of the following is free of non-determinism?

a. TG

b. NFA

c. NFA-^

d. FA

62. There is no question of accepting any language in:

a. FA

b. TG

c.GTG

d. Moore machine

63. A string will be accepted by an NFA if there exist_____one successful path.

a. Atmost

b. Maximum

c. atleast AL-JUNAID INSTITUTE OF GROUP

d. None of these

64. Kleene's theorem part I expresses the relationship between_____

a. FA and TG

b. TG and RE

c. RE and FA

d. FAand RE

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65. Keeping in view the discussion by Martin, how any states are required to recognize the language of all strings of length 2 or more defined over ∑={a,b} with 'b' being the second letter from right?

a. 6

b. 7

c. 8

d. 9

66. FA and _____ are same excepts that _____has unique symbol for each transition.

a. FA, TG

b. NFA, TG

c. GTG, NFA

d. NFA,FA

67. Subtraction of binary number is possible through.

a. Both complementing and incrementing machine

b. incrementing machine

c. complementing machine

d. Converting machine

68. Which of the following is the bypass and state elimination step in the context of Kleene's theorem part II proof?

a. 1

b. 2

c. 3

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d. 4

69. Keeping in view the discussion by Martin, how any states are required to recognize the language of all strings of length 2 or more defined over ∑={a,b} with 'b' being the second letter from right?

a.6

b. 7 AL-JUNAID INSTITUTE OF GROUP

c. 8

d.9

70.let L be the language of all strings. Defined over ∑ (0,1) ,ending in 10.which of the following strings

Are indistinguishable with respect to L with z being 0

a.100.101

b.111.101

c. 110 .101

d.010.101

71.IF r1 =(aa +bb)and r2 =(a+ b) then the language(aa +bb)(a+ b) will be generated by _____

a.(r1)(r2)

b.(r1 +r2)

c.(r2)(r1)

d.(r1)*

72. Indroducing new start state in case of multiple start states in the step no_____

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a.1

b.2

c.3

d.4

73 THE language having even number of a ‘ s and even number of b’ s defined over S=(a, b) is called____

a. EVEN-EVEN

B.ODD-ODD

c. PALINDROME

d. FACTORIAL

74 In NFA having no transition at certain state .FA can be build by introducing.

A .Empty state

b. Combination of states

c .initial state

d. final state

75 .for every three regular expressions R,S. and T. the languages denoted by R(S U T)and (RS)U(RT)are the_____

a .same

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b. Different

c. R(SUT)is greater

d. none of the given options

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76. Null strings can be specified on edges in:

a. Finite Automata

b Non-Deterministic Finite Automata

c. Melay Machine

d. Transition graph

77. What is false about the PLAINDROME LANGUAGE?

A Every word is reverse of itself.

B It is an infinite language.

C FA can be build for it.

D None of the given option

78. While finding RE corresponding to a TG, if TG has more than one start state then.

a Eliminate the old start state
b Replace the old start state with final state
c Replace the old final state with new start state

d. Introduce the new start state

79. All possible combination of string of a language including null string is referred as..

Kleene star closure of a languagec.

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80.N! will be equal to:

c. n*(n-1)!cfd

81.Every NFA can be considered to be a____ as well, but the converse may not be true.

a. TG

b. FA

c. RE

d. GTG

82.In proving Kleene theorem II , if three statuses are connected then middle state is removed btconnecting first and third and writing corresponding RE in:

a.

b.

c. Concentration

d.

83.In _____ there must be transition for all the letters of the string.

a. NFA

b.GTG

c TG

d. FA AL-JUNAID INSTITUTE OF GROUP

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84.For a given Moore Machine , the input string is '101010' , thus the output string would be of length:
a. |Input|
b. |Input-1|
c. Cannot be predicted
d. Length of input string+1

85.The FA can be drawn for the regular expression (a+b)* with minimum_____ state(s).
a. 1
b.2
c. 3
d. 4

86.Which of the following does not contribute while finding out the length of strings?
^ ANSEWR

87.The language of all strings defined over alphabet set ={x,y} that ends with same letters will have the maximum length of:
a. Finate
b. non of these
c. Infinite
d. Equal

88.Considering FA1 and FA2 having 2 states each. Now FA1+FA2 can have maximum_____ number of states.
a. 2
b. 3
c. More than 3
d. Non of them

89.Which one of the following is RE for the language defined over ∑={a,b} having all the words starting with a?

a(a+b)*

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90.An _____ can be considered to be an intermediate structure between finite automaton and transition graph.
a. RE
b. NFA
c. GTG
d. None

91.In order to make NFA for the union of FA1 and FA2, the new initial state should be liked to:
a. initial state of FA1 only
b. initial and final states of FA1 and FA2 respectively
c. final and initial states of FA1 and FA2 respectively
d. Initial states of both FAs

92.We cannot construct an NFA for the language of _____ defined over alphabet set{a,b}.
a. Palindromes
b. Even
c. Odd
d. Integers

93.The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by the …… production trees
a. one
b. Two
c. More then one / Different
d. At most one

94.What do automata mean?
a. something done manually
b. something that works automatically / "something are done automatically
c. something done automatically

95.According to theory of automata there
a. 1
b. 2
c. 3

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d. 4

96.The Minimum length of the string (except bull string) of a language that starts and ends in the same letters will be:

a.1

b. 2

c. 3

d. All of above

97. If S= {ab,bb} then S* will not contain_____

a. Babb

b. Bbb

c. Bbba

d abbb

98.Which of the following machines has only one initial state and no final state?

a. More machine

b. Finite state machine

c.Deterministic finite state machine

d.Non deterministic finite state machine

98.Which of the following machines has only one initial state and no final state?

a. More machine

b. Finite state machine

c.Deterministic finite state machine

d.Non deterministic finite state machine

99.Which of the following diagram is very rigid in order to express any language ?

a.TG

b.NFA-^

c.GTG

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d.FA

100.If S={a},then S+will be_____
a.{a,aa ,aaa , aaaa…}
b.{a,aa,aaa,aaaa…}

c.{a,aaa,aaaaa,aaaaaaa,…..}

d. {aa, aaaa,aaaaaa,aaaaaaaa,…}

101.Let L be the language of the all string defined over over∑ = {0,1} ending in10. Which of the following string are indistinguishable with respect to L with z being 11?
a.111,101
b.001,101
c.010,101
d.111,111
102.Melay machine can have _____final state
1) Zero
2) One
3) More than one but finite
4) More than one but infinite

103.Given the Language L = {ab, aa,baa},which of the following string are in L*?
5) Abaabaaaabaa
6) Aaaabaaaa
7) Baaaaabaaaaab
8) Baaaaabaa
a)1,2 and 3
b)2,3 and 4
c)1,2 and 4

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d)1,3 and 4

104.If L1 and L2 are regular Language L1∩ ($L2 \cup L1$) will be
a.non regular
b.may be regular
c. Regular
d.none of the mentioned
105.In Mealy Machine the out put depends on_____
a. state
b.previous state
c.state and input
d.only input
106.There is no question of accepting any language in.
a.Mealy machine
b. Moore Machine
c.Moore and mealy
d.none of these.ge
107.Thestate where there is no way to leave after entry is called___
a.Davey john Locker
b.Initial State

c.Final state

d.Non-final state

108.FA corresponding to an NFA can be built by introducing an empty state for a letter having
a.No transaction at certain state
b.More than one transition at certain state

c.None of the given option

109.Which of the following diagram express language more simply?
a.Tg
B.Gt
c.GTG
d.All of above
110.Automata is the plural of_____
a.Auto
b. Automation

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c.Meta
d.All of above

111.If A and B are regular Language !(A∪B) is
a.Regular
b.none regular
c.may be regular
d.none of the mentioned
112.In NFA having no transition at certain state FA can be built by introducing.
a. Empty state.
B.Final state
C.initial state
D none of these

113.Consuming FA1 and FA2 having 2 STATES each. Now FA1+FA2 can have maximum_____number of state
a.2
b.3
c. More then 3
d. None of the given
114.In an FA when there is no path starting from initial state and ending state in final state then that FA_____
a.Reject all
b. Accept all string
c.Reject and accept all
d.None of above
115.According to theory of automation there are_____type if language
a. Two
b.Three
c.One
d.Four
116.In Moore machine if the length of input string is 9 then the length of output string will be.
a.20
b.60
c.10

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d.40

117.When ODD language is expressed by an FA,then it will have minimum _____ state

a. One

b TWO

c THREE

d FOUR

118.[(a+b)(a+b)]*.given RE contact generate the string_____

a. aaabbbbccc

b. Bbbbbb

c. aabbcc

d aaabcbbcbacc

119.Which of the following state is true about GTG?

a. Transaction are based on regular expression

120.Every_____is a _____ as will, but the converts may not be true.

a. FA ,TG

b. TG RE

c. GTG

d TG

121.Which of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?

a.TG andRE

b.REand FA

c.FA and RE

d. Mealy Machine

122.The recursive method for defining a language has ____ steps

a. two

b. Three

c. four

d. five

123.Consider the following RE.

.A(a+b)b*

All of the following word are accepted except_____

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A Aa

b.Aabb
c.Bbbba
d.None of these

124.Which of the following regular expression represented same language?

1. (a+ab)*
2. 2(ba+a)*
3. A*(aa*b)*
4. (a*b)*

a.2 and 1
b.3 and 2
c.1 and 2
d.3 and 4

125 For every there regular expression R,S and T the Language denoted by R(SUT) and (RS) U(RT) are the

a.Different
b. Same
c.Equal
d.All of above

126 Alphabet S={a,bc,cc} has____number of letters.

a. two
b. three
c. four
d. five

127 An____can be considered to be an intermediate structure between Finite automation and Transition Graph.

a.FA
b.RE
cBoth a and b
d.NFA

128 Two FAs are said to be equivalent if they_____

a.Simple langue
b.Accept different language
c. Accept same language

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d. All of above

129 There may be more then one transition for a certain letter on a state in

a. Finite Automata

b.In finite automata

d. None of these

130 _____can also help in proving Kleene Theorem III.

A. RE

b.Rna

c.FA

d.NFA

131 Kleene Theorem Part II expression the relationship between____

a.TG and FA

b. TG and RE

c.TG and NFA

d.None of these

132 FA corresponding to an NFA can be built by introducing an empty state for a letter having.

a. No transition at certain state

b.Transition at certain state

c.Transition at final state

d. None of these

133 FA is also called

a.FA

b.RE

c. DFA

d.All of above

134 If two Res generated same language then these Res are called ____

a.Equivalent FA

b. Equivalent RE

c.Equivalent DFA

d.none of these

135 We cannot an NFA for the language of ____ defend over alphabet set{a,b}

Palindromes (**answer**)

136 Kleene Theorem Part II expression the relationship between____
a. RE and FA
b.DFA and RE
c.FA and DFA
d.None of these
137 Let FA3 an FA corresponding to FA1 FA2 then the initial state of FA3 must correspond to the initial state of
a.Dfa
b.FA2 only
c. FA1 only
ddnone of these
138 Every FA should be___
a.Simple
b.Accurate
c.Non-deterministic
d. Deterministic
139 The minimum length of string (except null string) of a language that starts and ends in the different letter will be:
a.2
b.3
c.4
d.1
140 Which of the following will be the final state of FA3 obtained from the union of FA1 and FA2?
a.Initial state
B.Final states of FA1 or FA2
c.Initial state of FA1 orFA 2
d.None of the given

141.In concatenation we accept the initial state of FA2 automatically after the final state of FA1 because of:
a.initial state

b.Final state

c.We need just one final state

d.We need just one initial state

142 Let FA1 accepts many strings and FA2 accepts none then FA1+FA2 will be equal to:

a. FA2-FA1 (ANSWER)

143 The language {a ab aba bab} is _____ .

a.Irregular

b.Normal

c. Regular

d.All of above

144 Decomposing a string into its valid units is referred as:

a.Customizing

b. Tokenizing

c. Decomposing

d. Splitting

145 Let FA3 be an FA corresponding to FA1+FA2, then the initial state of FA3Must corresponds to the initial state of

a.NFA

b.DFA

c.RE

d.FA1 or FA2

146 If FA1 corresponds to (a+b)* then FA1 must accept _____ string/strings. Select correct option: No Oddlength

a.Some

b. EVERY

c.Just one

d.None of these

147 A regular language can be:

a. irregular

b. infinite

c. non-deterministic

d. None of the given options

148 N into its valid units is referred as:

d. Tokenizing

149 The strings of FA2 are accepted first before the strings of FA1

a.FA

b.NA

c. Palindromes

d.Non of these

150 There _____ a language for which only FA can be built but not the RE.

a.Is

b.Are

c.Be may

d.None of above

151 Kleene's Theorem part I expresses the relationship between ___

a.TG and Na

b.NA and FA

c.TG and TG

d. FA and TG

152 We can create an equivalent ___ for a language for which we create an ___

a.NA

b.NFA, FA

c.RNA

d.None of these

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CS402 - Theory of Automata (Quiz# 01) **Quiz Start Time: 05:36 PM, 13 December 2022**

Question # 5 of 10 (Start time: 05:41:35 PM, 13 December 2022) **Total Marks:** 1

If r1 and r2 are regular expressions then (r1 * r2) is _____ .

Select the correct option

- FA
- GTG
- RE
- TG

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CS402 - Theory of Automata (Quiz# 02) Quiz Start Time: 06:09 PM, 22 December 2022

Question # 4 of 10 (Start time: 06:14:04 PM, 22 December 2022) Total Marks: 1

Decomposing a string into its valid units is referred as:

Select the correct option

- Decomposing
- Dividing
- Tokenizing
- Splitting

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CS402 - Theory of Automata (Quiz# 02) **Quiz Start Time: 06:09 PM, 22 December 2022**

Question # 2 of 10 (Start time: 06:11:24 PM, 22 December 2022) **Total Marks:** 1

In _____ there must be transitions for all the alphabets over which a language is defined.

Select the correct option

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- (●) FA
- ( ) GTG
- ( ) TG
- ( ) NFA

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Question # 5 of 10 (Start time: 06:14:53 PM, 22 December 2022) **Total Marks:** 1

If L1 and L2′ are regular languages, L1 ∩ (L2′ U L1′)′ will be

Select the correct option

- non regular
- none of the mentioned
- may be regular
- regular

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CS402 - Theory of Automata (Quiz# 02) **Quiz Start Time: 06:09 PM, 22 December 2022**

Question # 6 of 10 (Start time: 06:16:06 PM, 22 December 2022) **Total Marks:** 1

Moore machine can have -------- final states.

Select the correct option

- ○ One
- ○ More than one but finite
- ○ More than one but infinite
- ◉ Zero

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CS402 - Theory of Automata (Quiz# 02)

Quiz Start Time: 06:09 PM, 22 December 2022

Question # 10 of 10 (Start time: 06:21:03 PM, 22 December 2022)

Total Marks: 1

Moore Machine is an application of:

Select the correct option

- ○ None of the mentioned
- ○ Finite automata without input
- ○ Non- Finite automata with output
- ◉ Finite automata with output

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CS402 - Theory of Automata (Quiz# 02) **Quiz Start Time: 06:09 PM, 22 December 2022**

Question # 1 of 10 (Start time: 06:10:03 PM, 22 December 2022) **Total Marks:** 1

NFA corresponding to union of FAs is built by introducing a new start state and connect it to the states originally connected to the old start state with the --------- transitions as the old start state:

Select the correct option

- ○ Concatenated
- ◉ Same
- ○ Different
- ○ Union of

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CS402 - Theory of Automata (Quiz# 01) Quiz Start Time: 05:36 PM, 13 December 2022

Question # 3 of 10 (Start time: 05:38:57 PM, 13 December 2022) **Total Marks:** 1

If S = {aa, bb} then S* will not contain _____.

Select the correct option

- aabbaa
- aabbbb
- aaabbb
- bbaabbbb

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Today 402 time 10:30am

Aslam Alaikum Today my Cs402.paper

Mcqs from Handouts 2 3 from Junaid file

Question

1

St ;RE is given And Draw NFA

2

Nd ;one is Sigma is given and write RE

3

Rd;One prove that(S+)*= (S*)*

4

Th;One NfA digram is Given and write RE With All step

Paper was very easy

Learning With A&I

Paper No 2

2 Jan 2023

At 2:30

Mcqs bht easy thy past paper Mae Sy thy

Jo handouts Sy thy bohot easy thy

Short Question 3 marks

Tg draw krn tha 3 marks

Difference between ab and (a+b) expression 3marks

Long Question

The language can express in FA then why we need NFA. Justify your answer.(5 marks)

Aik Long Question Yaad nhi

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CS402:Grand Quiz

Quiz Start Time: 11:29 AM, 30 December 2020

Question # 1 of 30 (Start time: 11:29:54 AM, 30 December 2020)

Total Marks: 1

If L1 and L2′ are regular languages, L1 ∩ (L2′ U L1′)′ will be

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Select the correct option

- (●) regular
- ( ) non regular
- ( ) may be regular
- ( ) none of the mentioned

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CS402:Grand Quiz

Quiz Start Time: 11:29 AM, 30 December 2020

Question # 4 of 30 (Start time: 11:34:05 AM, 30 December 2020)

Total Marks: 1

In Mealy machine the output depends on _____

Select the correct option

- Only present state
- Present state and Present input
- Nothing
- Type of input

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Quiz Start Time: 11:29 AM, 30 December 2020

Question # 21 of 30 (Start time: 11:54:22 AM, 30 December 2020)

Total Marks: 1

There is no question of accepting any language in:

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Select the correct option

- FA
- Moore machine
- Melay machine
- Both moore and melay machine

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CS402:Grand Quiz

Quiz Start Time: 11:29 AM, 30 December 2020

Question # 25 of 30 (Start time: 12:00:13 PM, 30 December 2020)

Total Marks: 1

The state where there is no way to leave after entry is called _____.

Select the correct option

- (●) Davey John locker
- ( ) initial state
- ( ) final state
- ( ) non-final state

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CS402:Grand Quiz

Quiz Start Time: 12:22 PM, 30 December 2020

Question # 8 of 30 (Start time: 12:29:42 PM, 30 December 2020)

Total Marks: 1

FA corresponding to an NFA can be built by introducing an empty state for a letter having

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Select the correct option

- (●) no transition at certain state
- ( ) one transition at certain state
- ( ) two transitions at certain state
- ( ) more than two transitions at certain state

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CS402:Grand Quiz

Quiz Start Time: 11:29 AM, 30 December 2020

Question # 23 of 30 (Start time: 11:57:00 AM, 30 December 2020)

Total Marks: 1

Which of the following diagrams expresses languages more simply?

Select the correct option

- FA
- NFA
- TG
- GTG

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Quiz Start Time: 11:29 AM, 30 December 2020

Question # 28 of 30 (Start time: 12:03:39 PM, 30 December 2020)

Total Marks: 1

Automata is the plural of _____.

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Select the correct option

- automate
- automaton
- automation
- automatic

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Quiz Start Time: 11:29 AM, 30 December 2020

Question # 5 of 30 (Start time: 11:35:34 AM, 30 December 2020)

Total Marks: 1

If A and B are regular languages, !(A' U B') is:

Select the correct option

- (●) regular
- ( ) non regular
- ( ) may be regular
- ( ) none of the mentioned

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Quiz Start Time: 11:29 AM, 30 December 2020

Question # 12 of 30 (Start time: 11:44:27 AM, 30 December 2020)

Total Marks: 1

In NFA having no transition at certain state, FA can be built by introducing:

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Select the correct option

- (●) Empty state
- ( ) Combination of states
- ( ) Initial state
- ( ) Final state

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Quiz Start Time: 01:51 PM, 30 December 2020

Question # 2 of 30 (Start time: 01:52:39 PM, 30 December 2020)

Total Marks: 1

If S = { x }, then S* will be _____.

Select the correct option

- {^,x,xxx,xxxx,xxxxx,...}
- {^,x,xx,xxx,xxxx,...}
- {^,x,xxx,xxxxx,xxxxxxx,...}
- {^,xx,xxxx,xxxxxx,xxxxxxxx,...}

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CS402:Grand Quiz

Quiz Start Time: 11:29 AM, 30 December 2020

Question # 14 of 30 (Start time: 11:46:59 AM, 30 December 2020)

Total Marks: 1

Considering FA1 and FA2 having 2 states each. Now FA1+FA2 can have maximum _____ number of states.

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Select the correct option

- 2
- 3
- more than 3
- None of the given options

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Quiz Start Time: 01:51 PM, 30 December 2020

Question # 1 of 30 (Start time: 01:51:55 PM, 30 December 2020) **Total Marks: 1**

In an FA, when there is no path starting from initial state and ending in final state then that FA_____.

Select the correct option

- accept null string
- accept all strings
- accept all non empty strings
- does not accept any string

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Quiz Start Time: 12:22 PM, 30 December 2020

Question # 2 of 30 (Start time: 12:23:51 PM, 30 December 2020)

Total Marks: 1

According to theory of automata there are _____ types of languages.

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Select the correct option

- One
- Two
- Three
- Four

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Quiz Start Time: 12:22 PM, 30 December 2020

Question # 25 of 30 (Start time: 12:47:10 PM, 30 December 2020)

Total Marks: 1

In Moore machine, if the length of input string is 9, then the length of output string will be:

Select the correct option

- 7
- 8
- 9
- 10

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Quiz Start Time: 12:22 PM, 30 December 2020

Question # 23 of 30 (Start time: 12:45:41 PM, 30 December 2020)

Total Marks: 1

When ODD language is expressed by an FA, then it will have minimum _____ states.

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Select the correct option

- (●) one
- ( ) two
- ( ) three
- ( ) four

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Quiz Start Time: 11:29 AM, 30 December 2020

Question # 30 of 30 (Start time: 12:05:52 PM, 30 December 2020)

Total Marks: 1

[(a + b)(a + b)]*, given RE cannot generate the string _____.

Select the correct option

- abbaabab
- abbbaa
- bbbbbb
- abbbaaaaa

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Quiz Start Time: 01:51 PM, 30 December 2020

Question # 22 of 30 (Start time: 02:07:50 PM, 30 December 2020)

Total Marks: 1

Which of the following statement is true about GTG?

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Select the correct option

- Transitions are based on input letters
- Transitions are based on specified substrings
- Transitions are based on regular expressions
- Transitions are based on alphabet set

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CS402:Grand Quiz

Quiz Start Time: 01:51 PM, 30 December 2020

Question # 5 of 30 (Start time: 01:54:20 PM, 30 December 2020)

Total Marks: 1

Every _____ is a _____ as well, but the converse may not be true.

Select the correct option

- TG, FA
- FA, TG
- TG, GTG
- FA,GTG

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CS402:Grand Quiz

Quiz Start Time: 12:22 PM, 30 December 2020

Question # 18 of 30 (Start time: 12:40:12 PM, 30 December 2020)

Total Marks: 1

Which one of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?

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Select the correct option

- Moore machine
- Mealy machine
- Finite state machine
- Deterministic finite state machine

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CS402:Grand Quiz

Quiz Start Time: 01:51 PM, 30 December 2020

Question # 24 of 30 (Start time: 02:09:14 PM, 30 December 2020)

Total Marks: 1

The recursive method for defining a language has _____ steps.

Select the correct option

- one
- two
- three
- four

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CS402:Grand Quiz

Quiz Start Time: 01:51 PM, 30 December 2020

Question # 6 of 30 (Start time: 01:55:05 PM, 30 December 2020)

Total Marks: 1

Consider the following RE:
a(a + b)b*
All of the following words are accepted except _____.

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Select the correct option

- aab
- abb
- aa
- aba

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CS402:Grand Quiz

Quiz Start Time: 11:29 AM, 30 December 2020

Question # 29 of 30 (Start time: 12:05:01 PM, 30 December 2020)

Total Marks: 1

Which of the following regular expressions represent same language?

1. (a+ab)*
2. (ba+a)*
3. a*(aa*b)*
4. (a*b*)*

Select the correct option

- (●) 1 and 2
- ( ) 1 and 3
- ( ) 3 and 4
- ( ) 1 and 4

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CS402:Grand Quiz

Quiz Start Time: 01:51 PM, 30 December 2020

Question # 10 of 30 (Start time: 01:57:43 PM, 30 December 2020) **Total Marks: 1**

For every three regular expressions R, S, and T, the languages denoted by R(S U T) and (RS) U (RT) are the _____ .

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Select the correct option

- (●) Same
- ( ) Different
- ( ) R(S U T) is Greater
- ( ) None of the given options

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CS402:Grand Quiz

Quiz Start Time: 01:51 PM, 30 December 2020

Question # 27 of 30 (Start time: 02:10:29 PM, 30 December 2020)

Total Marks: 1

Alphabet S = {a, bc, cc} has _____ number of letters.

Select the correct option

- One
- Two
- Three
- Four

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CS402:Grand Quiz

Quiz Start Time: 11:29 AM, 30 December 2020

Question # 22 of 30 (Start time: 11:56:00 AM, 30 December 2020)

Total Marks: 1

An _____ can be considered to be an intermediate structure between Finite automaton and Transition Graph.

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Select the correct option

- RE
- GTG
- NFA
- None of the given options

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CS402:Grand Quiz

Quiz Start Time: 11:29 AM, 30 December 2020

Question # 2 of 30 (Start time: 11:31:21 AM, 30 December 2020)

Total Marks: 1

Two FAs are said to be equivalent, if they _____.

Select the correct option

- accept null string
- accept same language
- accept different language
- none of the given options

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CS402:Grand Quiz

Quiz Start Time: 12:22 PM, 30 December 2020

Question # 20 of 30 (Start time: 12:42:40 PM, 30 December 2020)

Total Marks: 1

There may be more than one transition for a certain letter on a state in:

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Select the correct option

- (●) Finite Automata
- ( ) Non-Deterministic Finite Automata
- ( ) Transition Table
- ( ) Moore Machine

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CS402:Grand Quiz

Quiz Start Time: 01:51 PM, 30 December 2020

Question # 7 of 30 (Start time: 01:55:32 PM, 30 December 2020)

Total Marks: 1

---------- can also help in proving Kleene Theorem III.

Select the correct option

- (●) NFA
- ( ) PDA
- ( ) Moore machine
- ( ) Melay machine

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CS402:Grand Quiz

Quiz Start Time: 01:51 PM, 30 December 2020

Question # 21 of 30 (Start time: 02:07:18 PM, 30 December 2020)

Total Marks: 1

Kleene's Theorem Part II expresses the relationship between _____.

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Select the correct option

- FA and TG
- TG and RE
- RE and FA
- FA and RE

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CS402:Grand Quiz

Quiz Start Time: 01:51 PM, 30 December 2020

Question # 14 of 30 (Start time: 02:01:23 PM, 30 December 2020) **Total Marks:** 1

When ODD language is expressed by an FA, then it will have minimum _____ states.

Select the correct option

- (●) one
- ( ) two
- ( ) three
- ( ) four

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CS402:Grand Quiz

Quiz Start Time: 01:51 PM, 30 December 2020

Question # 12 of 30 (Start time: 01:59:05 PM, 30 December 2020) Total Marks: 1

FA corresponding to an NFA can be built by introducing an empty state for a letter having

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Select the correct option

- (●) no transition at certain state
- ( ) one transition at certain state
- ( ) two transitions at certain state
- ( ) more than two transitions at certain state

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CS402:Grand Quiz

Quiz Start Time: 12:22 PM, 30 December 2020

Question # 26 of 30 (Start time: 12:48:01 PM, 30 December 2020)

Total Marks: 1

Let L be the language of all strings, defined over Σ = {0,1}, ending in 10. Which of the following strings are distinguishable with respect to L with z being 0?

Select the correct option

- (●) 111, 101
- ( ) 001, 101
- ( ) 111, 111
- ( ) 010, 101

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Quiz Start Time: 12:22 PM, 30 December 2020

Question # 3 of 30 (Start time: 12:24:29 PM, 30 December 2020)

Total Marks: 1

FA is also called

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Select the correct option

- TG
- GTG
- NFA
- DFA

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CS402:Grand Quiz

Quiz Start Time: 11:29 AM, 30 December 2020

Question # 13 of 30 (Start time: 11:45:49 AM, 30 December 2020) **Total Marks: 1**

If two RE's generate same language then these RE's are called _____.

Select the correct option

- Same RE
- Equal RE
- Similar RE
- Equivalent RE

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CS402:Grand Quiz

Quiz Start Time: 11:29 AM, 30 December 2020

Question # 24 of 30 (Start time: 11:59:08 AM, 30 December 2020)

Total Marks: 1

We cannot construct an NFA for the language of _____ defined over alphabet set {a,b}.

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Select the correct option

- Even even
- odd
- Palindromes
- Integers

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CS402:Grand Quiz

Quiz Start Time: 11:29 AM, 30 December 2020

Question # 18 of 30 (Start time: 11:52:03 AM, 30 December 2020)

Total Marks: 1

Kleene's Theorem Part III expresses the relationship between _____.

Select the correct option

- FA and TG
- TG and RE
- RE and FA
- FA and RE

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CS402:Grand Quiz

Quiz Start Time: 12:22 PM, 30 December 2020

Question # 1 of 30 (Start time: 12:22:39 PM, 30 December 2020)

Total Marks: 1

Let FA3 be an FA corresponding to FA1FA2, then the initial state of FA3 must correspond to the initial state of

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Select the correct option

- (●) FA1 only
- ( ) FA2 only
- ( ) FA1 or FA2
- ( ) FA1 and FA2

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CS402:Grand Quiz

Quiz Start Time: 11:29 AM, 30 December 2020

Question # 20 of 30 (Start time: 11:53:58 AM, 30 December 2020)

Total Marks: 1

Every FA should be_____.

Select the correct option

- (●) deterministic
- ( ) non-deterministic
- ( ) deterministic and non-deterministic
- ( ) not depends on language

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CS 402

Theory of Automata

Important Topics Midterm

- NFA
- Moore Machine
- Mealy Machine
- Kleen
- Regular Expression
- Finite and Infinite Language
- Finite and Infinite Languages

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- Final Term Short Notes
- Mid Term Past Paper
- Final Term Past Paper
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- Accurate Assignment
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MIDTERM EXAMINATION
Spring 2010
CS402- Theory of Automata

Question No: 1 (Marks: 1) - Please choose one
Σ={a,Aa,Abb}, then string aAaAbbAa has _____ length.

- ► One
- ► Two
- ► Three
- ► **Four (Page 4)**

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Question No: 2 (Marks: 1) - Please choose one
Languages generated by kleene star are always _____.

- ► Finite
- ► **Infinite (Page 7)**
- ► Sometimes finite & sometimes infinite
- ► None of the these

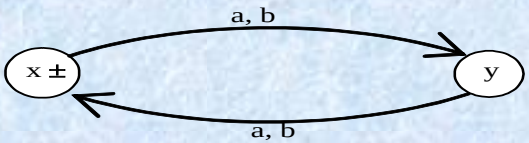
Question No: 3 (Marks: 1) - Please choose one
Let S = {aa, bb} be a set of strings then s* will have

- ► **Λ (Page 7)**
- ► abba
- ► aabbbaa
- ► bbaab

Question No: 4 (Marks: 1) - Please choose one
If r1 = (aa + bb) and r2 = (a + b) then the language (aa + bb)* will be generated by

- ► (r1)(r2)
- ► (r1 + r2)
- ► (r2)*
- ► **(r1)* (Page 10)**

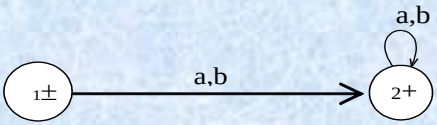
Question No: 5 (Marks: 1) - Please choose one



Above given FA can be represented by

- **((a+ b) (a + b))*** **(Page 13)**
- (a + b)(a + b)*
- (a + b)(a + b)
- (a + b)*(a + b)*

Question No: 6 (Marks: 1) - Please choose one



Above given FA accepts _____ strings defined over Σ={a , b}

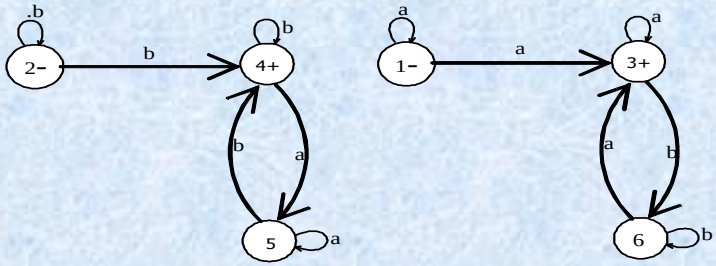
- **All** **(Page 15)**
- Some
- All but not null
- None of these

Question No: 7 (Marks: 1) - Please choose one

If a language can be expressed through FA, then it can also be expressed through TG.

- **True** **(Page 25)**
- False
- Depends on language
- None of the above

Question No: 8 (Marks: 1) - Please choose one

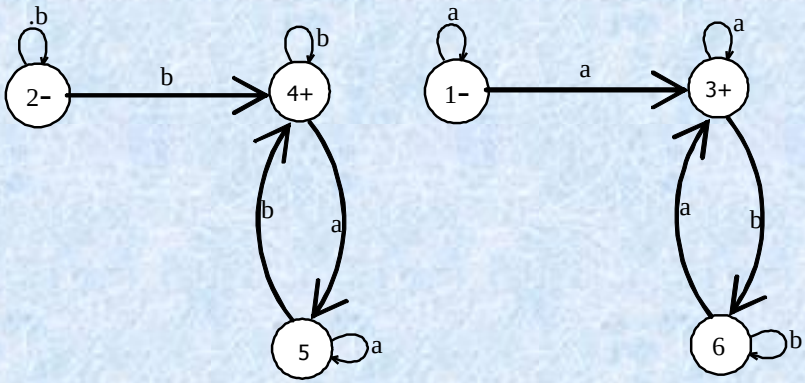


Above given TG has _____ RE.

- a+b+a(a + b)*a + b(a + b)*b
- a(a + b)*a + b(a + b)*b
- both are given
- **none of the given**

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Question No: 9 (Marks: 1) - Please choose one



Above given FA accepts the language in which strings

- ► **Begins with and ends in same letter**
- ► Begins with and ends in different letter
- ► Has length more than 2
- ► None of the given

Question No: 10 (Marks: 1) - Please choose one

GTG can have _____ final state.

- ► 0
- ► 1
- ► More than 1
- ► **All of the given** **Click here for detail**

Question No: 11 (Marks: 1) - Please choose one

In GTG, if a state has more than one incoming transitions from a state. Then all those incoming transitions can be reduced to one transition using _____ sign

- ► -
- ► + **(Page 27)**
- ► *
- ► None of the given

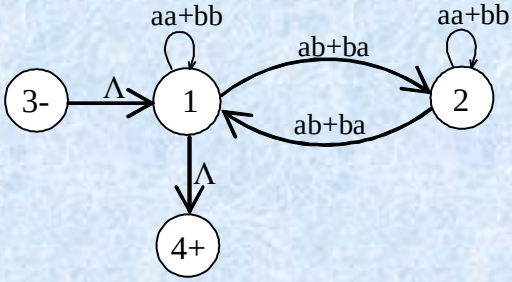
Question No: 12 (Marks: 1) - Please choose one

"One language can be expressed by more than one NFA". This statement is _____.

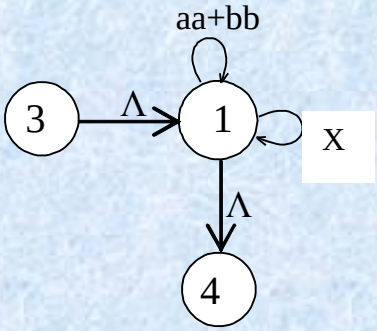
- ► False
- ► **True** **(Page 41)**
- ► Depends on NFA
- ► None of the given

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Question No: 13 (Marks: 1) - Please choose one



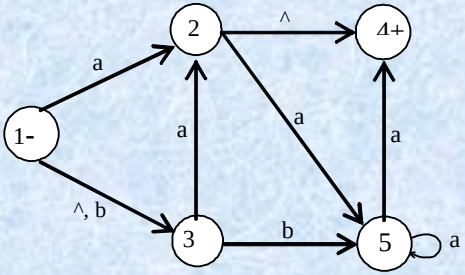
If above given TG is drawn like



Then what will be written in place of X.

- (ab+ba)(aa+bb)(ba+ab)
- (ab+ba)(aa+bb)(ab+ba)
- **(ab+ba)(aa+bb)*(ab+ba)** **(Page 31)**
- (ab+ba)(aa+bb)(ab+ba)*

Question No: 14 (Marks: 1) - Please choose one



Above given structure is an _____.

- FA
- NFA
- **NFA -^** **(Page 42)**
- TG

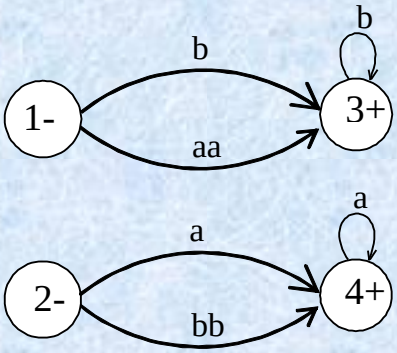
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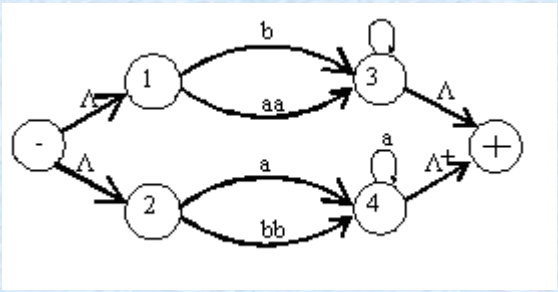
Question No: 15 (Marks: 1) - Please choose one

One FA has 3 states and 2 letters in the alphabet. Then FA will have _____ number of transitions in the diagram

- ► 4
- ► 5
- ► 7
- ► **6 (Page 14)**

Question No: 16 (Marks: 1) - Please choose one





Above given two TG's are _____.

- ► **Equivalent (Page 26 - 27)**
- ► None-equivalent
- ► Not valid
- ► None of the given

MIDTERM EXAMINATION
Spring 2010
CS402- Theory of Automata

Question No: 1 (Marks: 1) - Please choose one

If an alphabet has n number of letter, then number of strings of length m will be

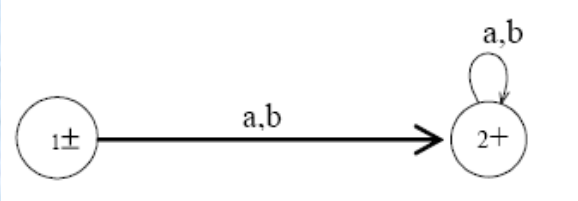
- ►n+m
- ► (n)(m)
- ►m^n
- ►**n^m (Page 6)**

ایماندار کو غصہ دیر سے آتا ہے اور جلدی دور ہو جاتا ہے

Question No: 2 (Marks: 1) - Please choose one
Languages generated by kleene star are always _____.
►Finite
►**Infinite (Page 7) rep**
►Sometimes finite & sometimes infinite
►None of the these

Question No: 3 (Marks: 1) - Please choose one
$1^*(1 + \Lambda) = 1^*$ this statement is
►**True**
►False
►Sometimes true & sometimes false
►None of these

Question No: 4 (Marks: 1) - Please choose one
a*b* = (ab)* this expression is _____
►True
►**False**
►Can't be assumed
►None of these

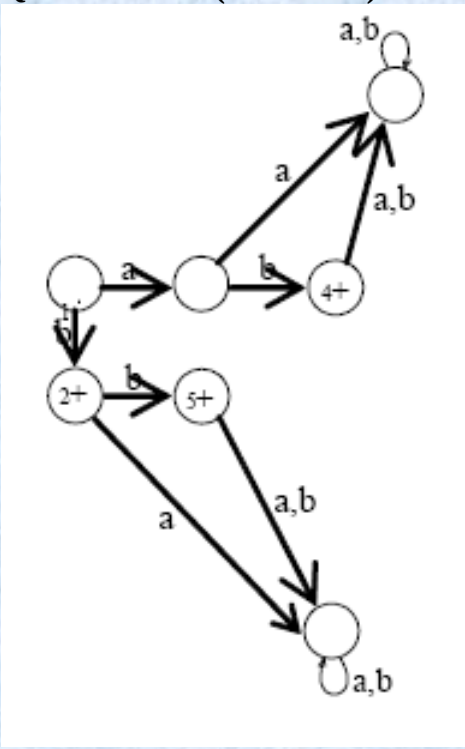
Question No: 5 (Marks: 1) - Please choose one


Above given FA can be expressed as _____
► **(a + b)* (Page 19)**
►a* + b*
► (ab + ba)*
►None of these

Question No: 6 (Marks: 1) - Please choose one
If a language is expressed through TG, then that language will have its RE.
►**True (Page 25)**
►False
►Depends on language
►None of these

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Question No: 7 (Marks: 1) - Please choose one



Above given FA accepts _____ language.

►**Finite (Page 17)**

►Infinite

►Depends on alphabet

►None of these

Question No: 8 (Marks: 1) - Please choose one

In TG there may exist more than one path for certain string.

►**True (Page 25)**

►False

►Depends on the language

►None of these

Question No: 9 (Marks: 1) - Please choose one

In TG there may exist no paths for certain string.

►**True (Page 25)**

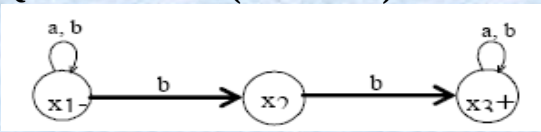
►False

►Depends on the language

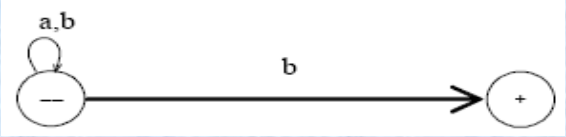
►None of these

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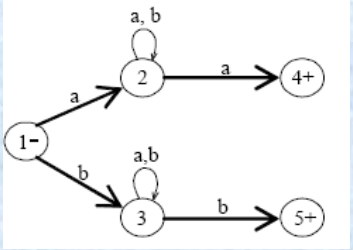
Question No: 10 (Marks: 1) - Please choose one
GTG can have _____ final state.
►0
►1
►More than 1
►**All of the given** Click here for detail Rep

Question No: 11 (Marks: 1) - Please choose one


Above given diagram is an NFA. If we convert it into an FA using transition table, then new FA will be consisted on _____ number of states.
►6
►5
►**4 (Page 45)**
►3

Question No: 12 (Marks: 1) - Please choose one


Above given TG accepts the language in which all strings
►**Ends in b**
►Begins with b
►Ends and begins with b
►None of the given

Question No: 13 (Marks: 1) - Please choose one


Above given TG has _____ RE.
►a(a + b)*b + b(a + b)*a
►b(b + a)*a + b(a + b)*a
►None of these
►**a(a + b)*a + b(a + b)*b**

جھوٹ انسان اور ایمان دونوں کا دشمن ہے

Question No: 14 (Marks: 1) - Please choose one
Every FA should be _____
►**Deterministic (Page 25)**
►Non- Deterministic
►Deterministic & Non- Deterministic
►None of these

MIDTERM EXAMINATION
Fall 2010
CS402- Theory of Automata

Question No: 1 (Marks: 1) - Please choose one
Auto Meta mean
► Manual work
► **Automatic work (Page 3)**

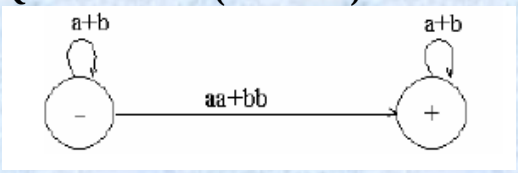
Question No: 2 (Marks: 1) - Please choose one
S= {a,bc,cc} has the latters
► 1
► 2
► **3**
► 4

Question No: 3 (Marks: 1) - Please choose one
S={a,bb,bab,baabb} set of strings then S* will not have
► **baba**
► baabbab
► bbaaabb
► bbbaabaabb

Question No: 4 (Marks: 1) - Please choose one
One language can represents more than one RE.
► **True (Page 9)**
► Falss
► Can't be assumed
► Non of given

عقل مند کہتا ہے میں کچھ نہیں جانتا جبکہ بے وقوف کہتا ہے کہ میں سب کچھ جانتا ہوں

Question No: 5 (Marks: 1) - Please choose one

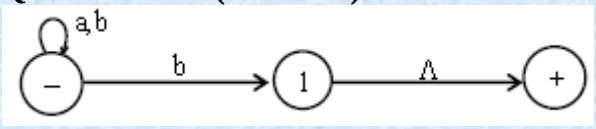


Given GTG has RE

► **(a+b)* (aa+bb)(a+b)*** **(Page 24)**

► None of option

Question No: 6 (Marks: 1) - Please choose one



NFA accept _____String

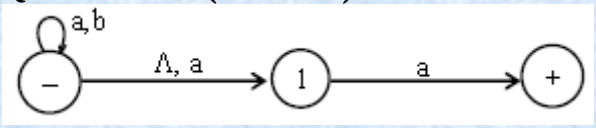
► b

► babab

► baaab

► **all** **(Page 43)**

Question No:7 (Marks: 1) - Please choose one



NFA accept _____String

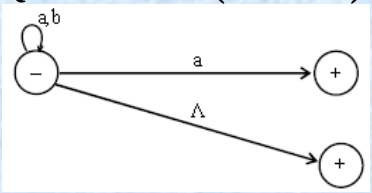
► bab

► a

► aba

► **a & aba** **(Page 43)**

Question No: 8 (Marks: 1) - Please choose one



TG has

► (a+b)*

► **Λ+(a+b)*a** **(Page 20)**

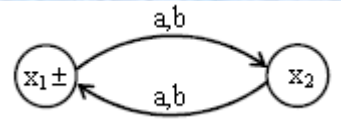
► Λ+(a+b)*a*

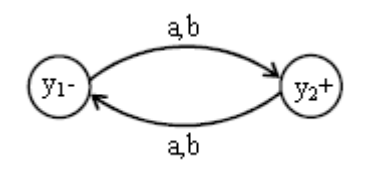
► None of given

خود کو تمھیں سے بڑھ کر کوئی اچھا مشورہ نہیں دے سکتا

Question No: 9 (Marks: 1) - Please choose one
TG can more then one initial state
► **True (Page 26)**
► False
► Depend on alphabets
► None of given

Question No:10 (Marks: 1) - Please choose one





RE will be
► (a+b)*
► (a+b)*(a*+b*)
► **None of the given (Page 37)**
Ref:- regular expression corresponding to above two FA's can be (a+b)((a+b)(a+b))* or ((a+b)(a+b))*(a+b)

Question No: 11 (Marks: 1) - Please choose one
The clouser FA*(star on an FA) always accept _____string
► **Null (Page 7)**
► aa
► bb
► None of given

Question No: 12 (Marks: 1) - Please choose one
In FA final state represent by _____sign
► +
► -
► =
► *

Question No: 13 (Marks: 1) - Please choose one
In FA one enter in specific stat but there is no way to leave it then state is called
► Dead States
► Waste Baskets
► Davey John Lockers
► **All of above (Page 17)**

جو لوگوں کے سامنے فخر کرتا ہے وہ لوگوں کی نظروں سے گر جاتا ہے

Question No: 14 (Marks: 1) - Please choose one

Using tree structure final state represent by

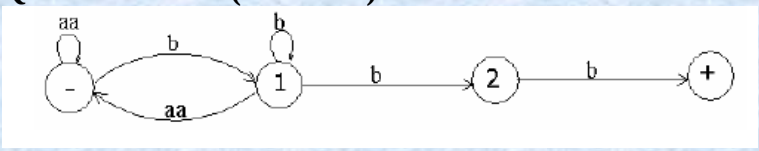
► *

► -

► **double circle (Page 13)**

► None of given

Question No: 15 (Marks: 1) - Please choose one



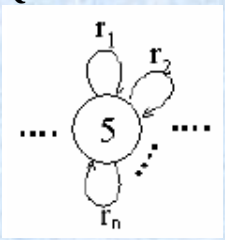
► **a's occur only in even clumps and that ends in three or more b's (Page 22)**

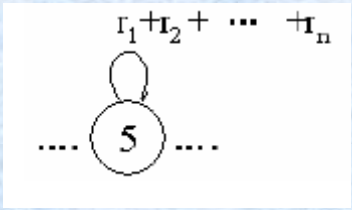
► length larger then 2

► it does not accept any language

► none of given option

Question No: 16 (Marks: 1) - Please choose one





These GTG are _____

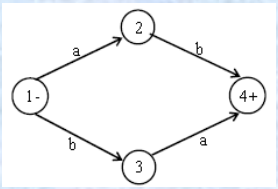
► **Equal (Page 27)**

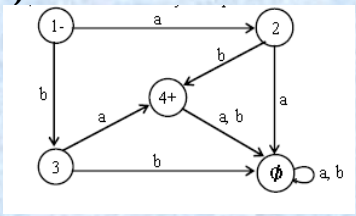
► Not equal

► Not valid

► None of given

Question No: 17 (Marks: 1) - Please choose one





NFA to FA will_____

► **Equal (Page 43)**

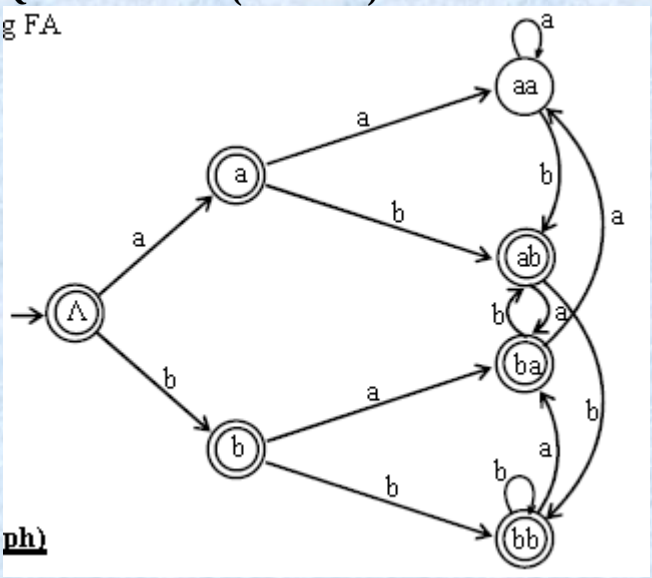
► Not equal

► Not valid

► None of given

جو شخص ناکامیوں سے ڈر کر بھاگتا ہے کامیابی اُس سے ڈر کر بھاگتی ہے

Question No: 18 (Marks: 1) - Please choose one



FA having RE_____

Λ + a + b + (a+b)*(ab+ba+bb) (Page 18)

a + b + (a+b)*(ab+ba+bb).

MIDTERM EXAMINATION
Fall 2010
CS402- Theory of Automata

Question No: 1 (Marks: 1) - Please choose one

Length of strings, generated by infinite language is_____

►**finite (Page 7)**

►infinite

►none of these

Ref:- By infinite language, it is supposed that the language contains infinite many words, each of finite length

Question No: 2 (Marks: 1) - Please choose one

RE for the language defined over Σ={a,b} having words starting with a is____

►**a(a+b)* (Page 12)**

► (a+b)*a

► (a+b)*

►None of these

بدصورت چہرہ بدصورت دماغ سے بہتر ہے

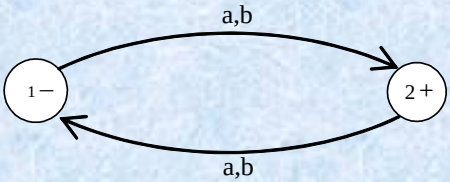
Question No: 3 (Marks: 1) - Please choose one

"Every Infinite language is regular" this statement is

►**True (Page 11)**

►False

Question No: 4 (Marks: 1) - Please choose one

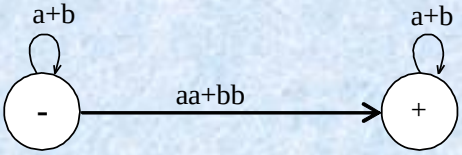


Above given FA accepts the null string

►True

►**False**

Question No: 5 (Marks: 1) - Please choose one



Above given GTG accepts the language in which strings

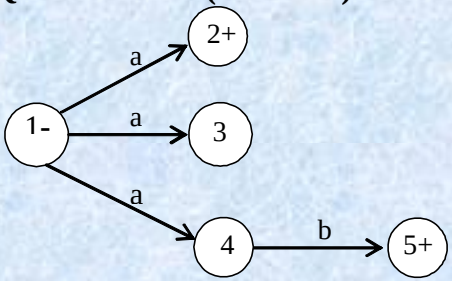
Contains double a or double b (Page 24)

Contains both a and double b

Depends on the alphabet

None of these

Question No: 6 (Marks: 1) - Please choose one



Above given NFA accepts the _____ number of strings

►one

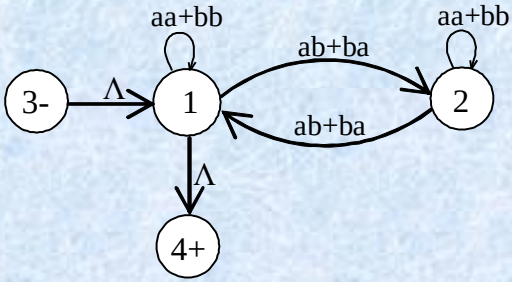
►**two (Page 40)**

►three

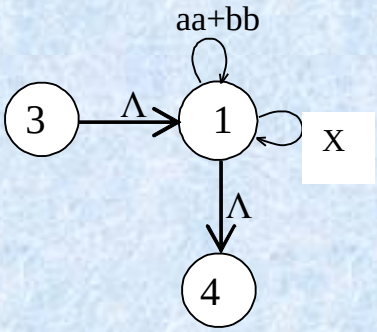
►four

عقل مند اپنے عیب خود دیکھتا ہے اور بیوقوفوں کے عیب دنیا دیکھتی ہے

Question No: 7 (Marks: 1) - Please choose one



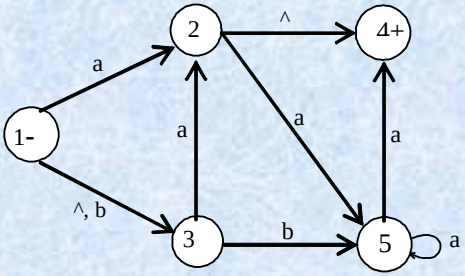
If above given TG is drawn like



Then what will be written in place of X.

- (ab+ba)(aa+bb)(ba+ab)
- (ab+ba)(aa+bb)(ab+ba)
- **(ab+ba)(aa+bb)*(ab+ba) (Page 31) rep**
- (ab+ba)(aa+bb)(ab+ba)*

Question No: 8 (Marks: 1) - Please choose one



Above given structure is an _____.

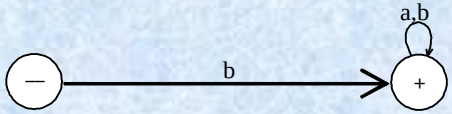
- FA
- NFA
- **NFA -^ (Page 42) rep**
- TG

عقل مند آدمی اس وقت تک نہیں بولتا جب تک خاموشی نہیں ہو جاتی

Question No: 9 (Marks: 1) - Please choose one
TG is always deterministic.
►True
►**False (Page 25)**

Question No: 10 (Marks: 1) - Please choose one
The length of output string in case of _____ is one more than the length of corresponding input string.
►**Finite Automaton (Page 55)**
►TG
►GTG

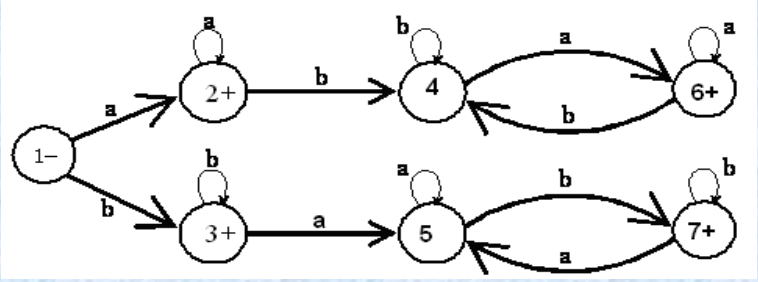
Question No: 11 (Marks: 1) - Please choose one



Above given TG accepts the _____ string.
►bb
►baba
►bbba
►**all of the given options (Page 19)**

Question No: 12 (Marks: 1) - Please choose one
If in an NFA, ∧ is allowed to be a label of an edge then that NFA is called _____.
Will not remain NFA
NFA with
NFA with null string (Page 42) rep
Either "NFA with null string" OR "NFA with "

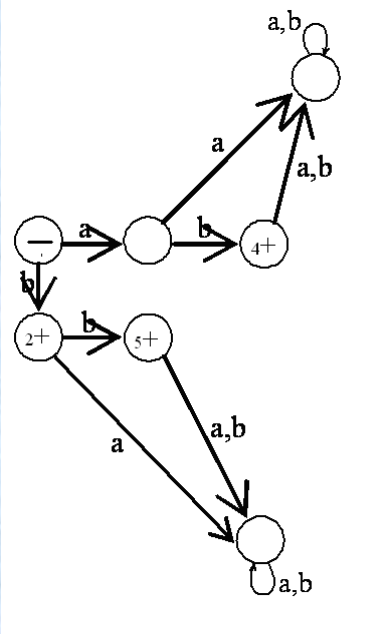
Question No: 13 (Marks: 1) - Please choose one



Above given FA generates the strings which _____
►**Starting and ending with same letters**
► Starting and ending with different letters
►None of these

بہترین تجربہ وہ ہے جس سے نصیحت حاصل ہو

Question No: 14 (Marks: 1) - Please choose one



Above given FA accepts _____ language.

►**Finite (Page 17) rep**

►Infinite

►Depends on alphabet

►None of these

MIDTERM EXAMINATION

Spring 2009

CS402- Theory of Automata (Session - 3)

Question No: 1 (Marks: 1) - Please choose one

Length of null string is

►Always not equal to 0

►**Always equal to 0** click here for detail

►It has variable length

►All are true

انسان دکھ نہیں دیتے بلکہ انسانوں سے وابستہ امیدیں دکھ دیتی ہیں

Question No: 2 (Marks: 1) - Please choose one
If an alphabet has n number of letter, then number of strings of length m will be
►n+m
► (n)(m)
►m^n
►n^m (Page 6) rep

Question No: 3 (Marks: 1) - Please choose one
Languages generated by kleene star are always _____.
►Finite
►Infinite (Page 7) rep
►Sometimes finite & sometimes infinite
►None of the these

Question No: 4 (Marks: 1) - Please choose one
"Every finite language can be expressed by FA". This statement is _____.
►True
►False
►Depends on language
►None of these

Question No: 5 (Marks: 1) - Please choose one
In FA, if one enters in a specific state but there is no way to leave it, then that specific state is called
►Dead States
►Waste Baskets
►Davey John Lockers
►All of these (Page 17)

Question No: 6 (Marks: 1) - Please choose one
In TG there may exist no paths for certain string.
►True (Page 25) rep
►False
►Depends on the language
►None of these

Question No: 7 (Marks: 1) - Please choose one
In GTG's there may exist no path for a certain string.
►True (Page 25)
►False
►Depends on alphabet
►None of these

فتنہ انگیز سچائی سے مصلحت آمیز جھوٹ بہتر ہے

Question No: 8 (Marks: 1) - Please choose one

In drawing FA3 (which is equal to FA1 + FA2), a state will be declared final if

►States of both FA's are final

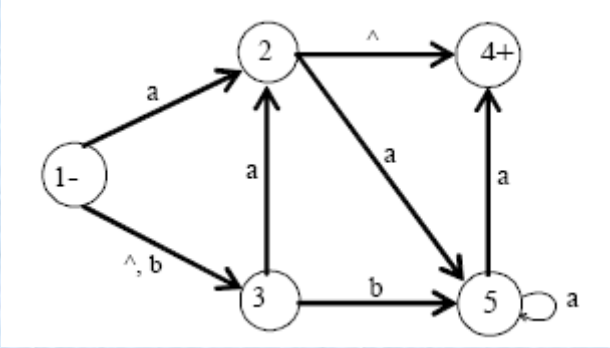
►At least one state is final (Page 32)

►Depends on language

►None of the given

Ref:- Let FA3 be an FA corresponding to r1r2, then the initial state of FA3 must correspond to the initial state of FA1 and the final state of FA3 must correspond to the final state of FA2.

Question No: 9 (Marks: 1) - Please choose one



Above given structure is an _____.

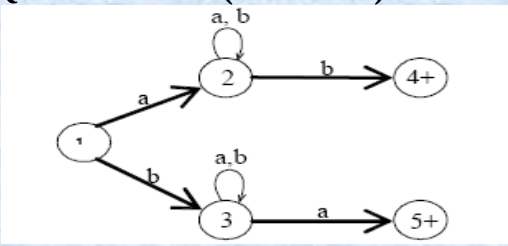
►FA

►NFA

►NFA -^ (Page 42) rep

►TG

Question No: 10 (Marks: 1) - Please choose one



►Above given TG represents the language____

►Begins and ends with same letters

►Begins and ends with different letters (Page 21)

►Begins with a

►None of these

خوبصورتی علم وادب سے ہوتی ہے لباس و حسن سے نہیں

Question No: 11 (Marks: 1) - Please choose one
In TG, there may be a transition for null string.
►**True (Page 18)**
►False
►Can't show transition for string
►None of these
Ref:- Finite set of transitions that show how to go from one state to another based on reading specified substrings of input letters, possibly even the null string (Λ).

Question No: 12 (Marks: 1) - Please choose one
The _____ machine helps in building a machine that can perform the addition of binary numbers.
►**Incrementing (Page 60)**
►Complementing
►Decrementing
►None of the given

Question No: 13 (Marks: 1) - Please choose one
►GTG can have _____ initial state.
►Zero
►One
►More than One
►**One OR more than One (Page 23)**
Ref:- Finite number of states, at least one of which is start state and some (maybe none) final states.

Question No: 14 (Marks: 1) - Please choose one
One FA has n states and m letters in the alphabet. Then FA will have _____ number of transitions in the diagram.
► (n)+(m)
► **(m)(n) OR (n)(m)**
►None of the given options
► (m)-(n)

Question No: 15 (Marks: 1) - Please choose one
If L1 and L2 are expressed by regular expressions r1 and r2, respectively then the language expressed by r1 + r2 will be _____
►**Regular (Page 10)**
►Ir-regular
►Can't be decided
►Another Language which is not listed here

تم اچھا کرو زمانہ تم کو برا سمجھے یہ اس سے بہتر ہے کہ تم برا کرو اور زمانہ تم کو اچھا سمجھے

Question No: 16 (Marks: 1) - Please choose one
Which statement is true?
►**All words are strings (Page 3)**
►All strings are words
►Both are always same
►None of these

MIDTERM EXAMINATION
Spring 2009
CS402- Theory of Automata (Session - 3)

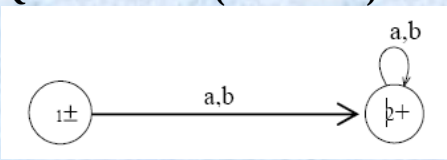
Question No: 1 (Marks: 1) - Please choose one
Alphabet S = {a,bc,cc} has _____ number of letters.
►One
►Two
►**Three rep**
►Four

Question No: 2 (Marks: 1) - Please choose one
One language can be represented by more than one RE" this statement is____
►False
►**True (Page 9)**
►Can't be assumed
►None of these

Question No: 3 (Marks: 1) - Please choose one
(a + b)*b is RE for the language defined over S={a,b} having words not ending in a
►**True (Page 13)**
►False
►Such a language is not regular
►None of these

انسان کے لئے بری صحبت سے بڑھ کر بری کوئی چیز نہیں

Question No: 4 (Marks: 1) - Please choose one



Above given FA accepts _____ strings defined over S={a , b}

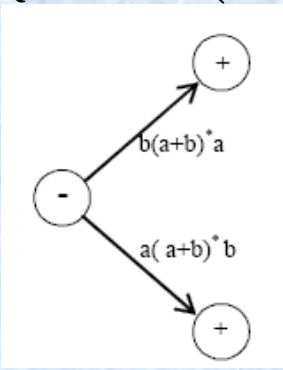
►**All (Page 15) rep**

►Some

►All but not null

►None of these

Question No: 5 (Marks: 1) - Please choose one



Above given GTG accepts the language in which strings

►**Begins and ends with different letters (Page 24)**

►Begins and ends with same letters

►Have length greater than 1

►None of these

Question No: 6 (Marks: 1) - Please choose one

According to 3rd part of the Kleene's theorem, If a language can be accepted by an RE then it can be accepted by a _____ as well

►TG

►FA

►**G and FA (Page 25)**

►None of these

Question No: 7 (Marks: 1) - Please choose one

If FA1 accepts no string and FA2 accepts many strings, then FA1 + FA2 will be equal to

►FA1

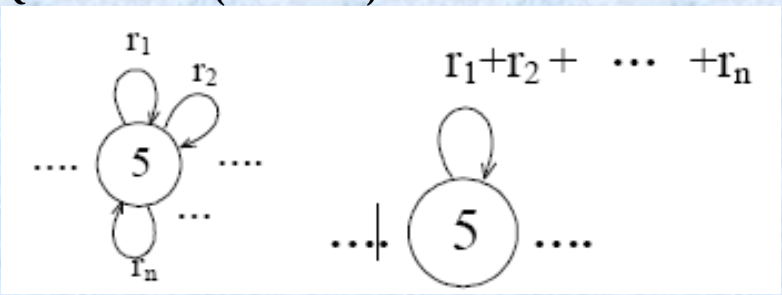
►**FA2**

►May be both

►None of the given

خاموشی غصے کا بہترین علاج ہے

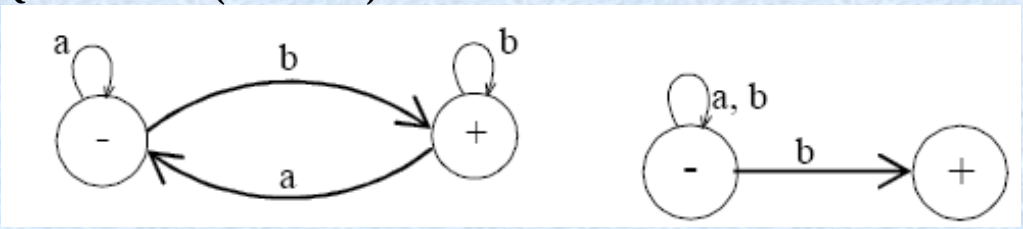
Question No: 8 (Marks: 1) - Please choose one



Above given GTG's are _____

►**Equivalent (Page 27) rep**
►Non-equivalent
►Non-valid
►None of the given

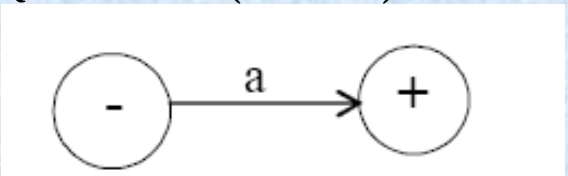
Question No: 9 (Marks: 1) - Please choose one



Above given NFA and FA generate same language.

►**True (Page 41)**
►False
►FA & NFA can't be equivalent
►None of these

Question No: 10 (Marks: 1) - Please choose one

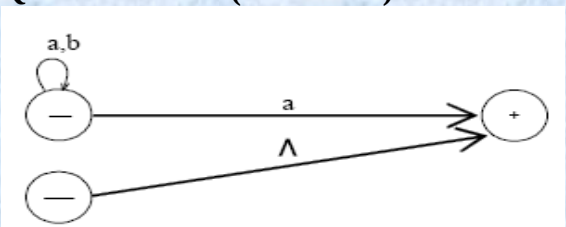


Above given structure is a _____

►FA
►TG
►**NFA (Page 44)**
►FA and NFA

افضل انسان وہ ہے جو اپنی اصلاح کی کوشش کرتا ہے

Question No: 11 (Marks: 1) - Please choose one



Above given TG has the _____ RE.

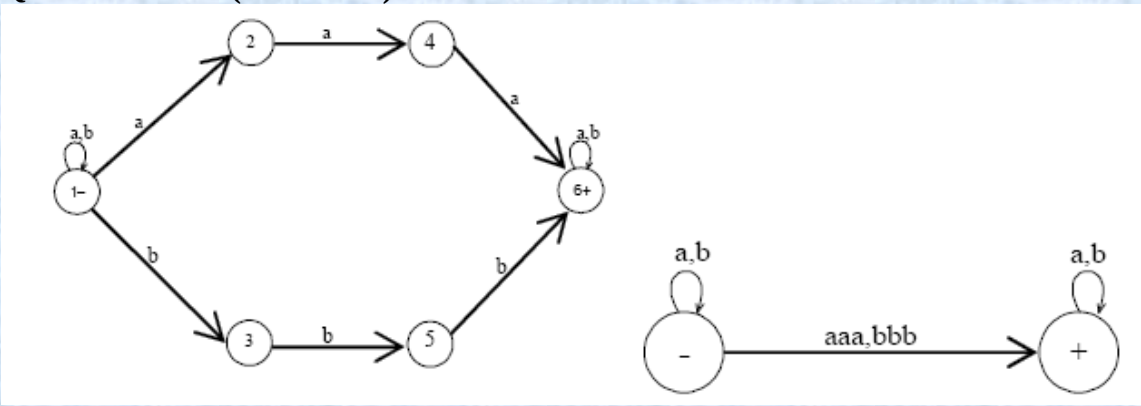
► (a + b)*a

►**λ+ (a + b)*a (Page 20)**

►None of these

►λ + (a + b)*a*

Question No: 12 (Marks: 1) - Please choose one



Above given TG's are _____.

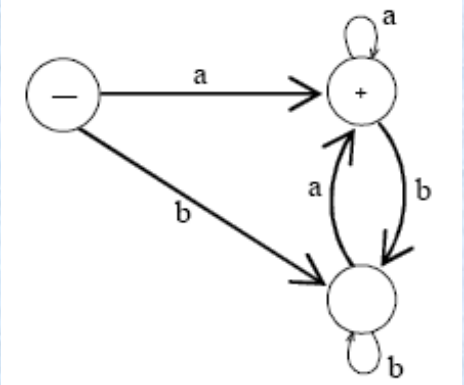
►**Equivalent (Page 21)**

►Non-equivalent

►TG's are not valid

►None of these

Question No: 13 (Marks: 1) - Please choose one



Above given FA has _____ RE.

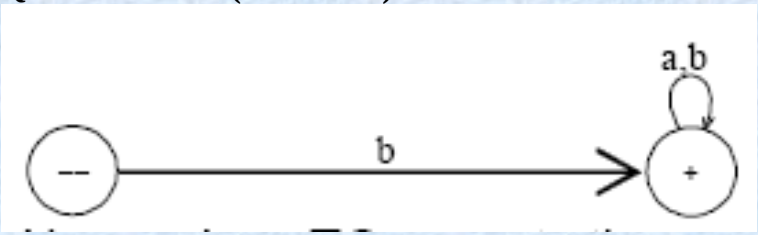
► **(a + b)*a (Page 13)**

► a(a + b)*

► ((a + b)*a)*

► (a + b)*a & ((a + b)*a)*

Question No: 14 (Marks: 1) - Please choose one



Above given TG accepts the _____ string.

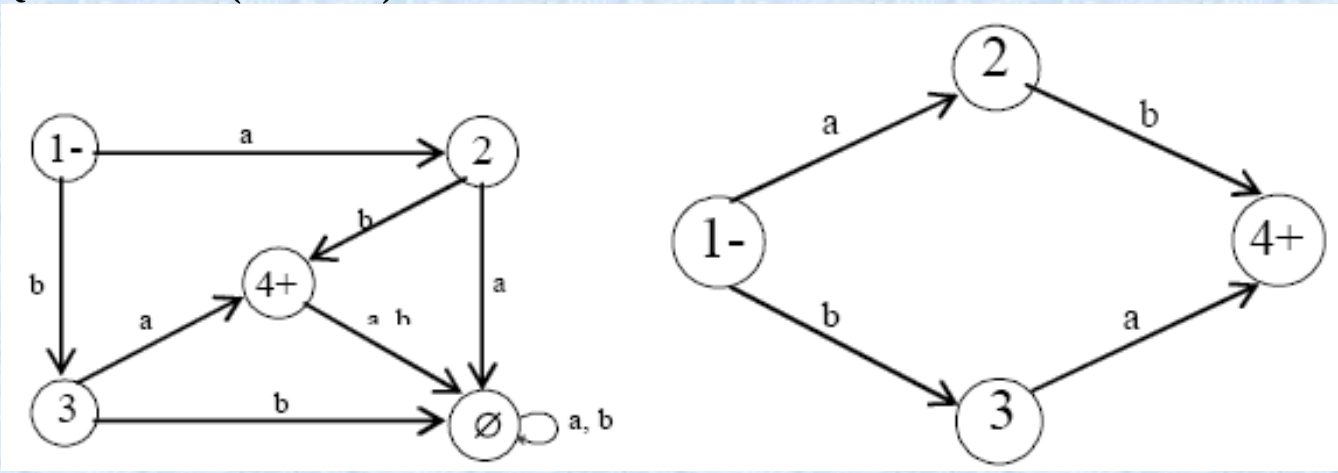
►bb

►baba

►bbba

►**all of the given options (Page 19) rep**

Question No: 15 (Marks: 1) - Please choose one



Above given FA and NFA are equivalent. This statement is _____.

►**True (Page 43) rep**

►False

►FA & NFA can never be equivalent

►None of the given options

جھوٹ رزق کو کھا جاتا ہے

MIDTERM EXAMINATION
Spring 2009
CS402- Theory of Automata (Session - 3)

Question No: 1 (Marks: 1) - Please choose one
If r1 and r2 are regular expressions then which of the following is not regular expression.
►r1 = r2
►r1r2
►r1*
►**r1 – r2 (Page 9)**

Question No: 2 (Marks: 1) - Please choose one
Which of the following is not a word of language EQUAL?
►aaabbb
►abbbabaa
►**abababa (Page 5)**
►bbbaaa

Question No: 3 (Marks: 1) - Please choose one
If S = {aa, bb}, then S* will not contain..
►aabbaa
►bbaabbbb
►**aaabbb**
►aabbbb

Question No: 4 (Marks: 1) - Please choose one
One language can be represented by more than one RE" this statement is____
►False
►**True (Page 9) rep**
►Can't be assumed
►None of these

Question No: 5 (Marks: 1) - Please choose one
"Every Infinite language is regular" this statement is
►**True (Page 11) rep**
►False
►Can't be supposed
►None of these

اطمینان قلب چاہتے ہو تو حسد سے دور رہو

Question No: 6 (Marks: 1) - Please choose one

PALINDROME can be defined by more than one regular language

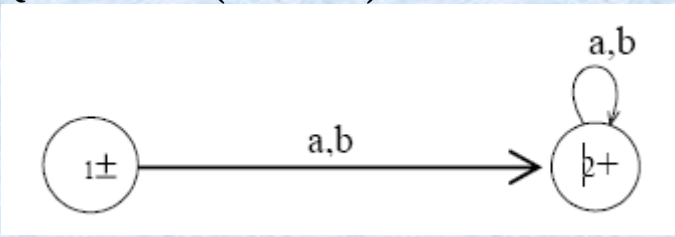
►True

►**False (Page 71)**

►By only one RE

►Some times By only one RE and Some times False

Question No: 7 (Marks: 1) - Please choose one



Above given FA can be expressed as _____

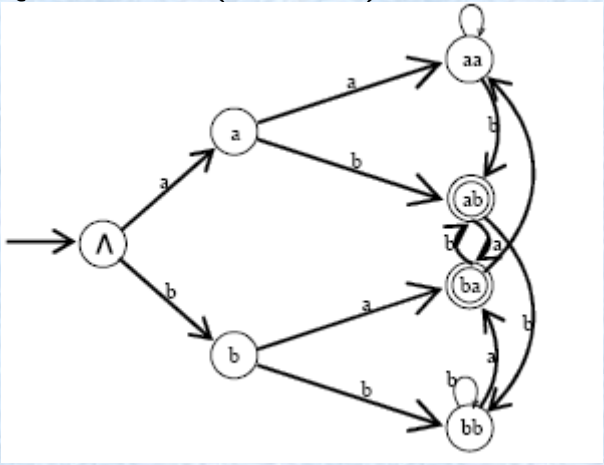
► **(a + b)* (Page 19) rep**

►a* + b*

► (ab + ba)*

►None of these

Question No: 8 (Marks: 1) - Please choose one



Above given FA is drawn using

►**Tree structure (Page 18)**

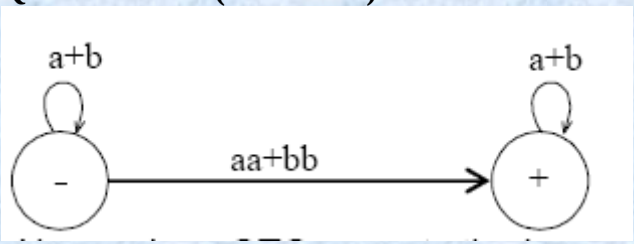
►It is not an FA

►Graph structure

►None of these

اس سے پہلے کہ تمہیں شہوت فتنے میں ڈالے نکاح کرلو

Question No: 9 (Marks: 1) - Please choose one



Above given GTG accepts the language in which strings

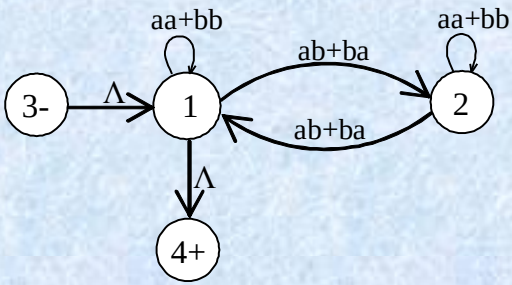
►**Contains double a or double b (Page 24) rep**

►Contains both a and double b

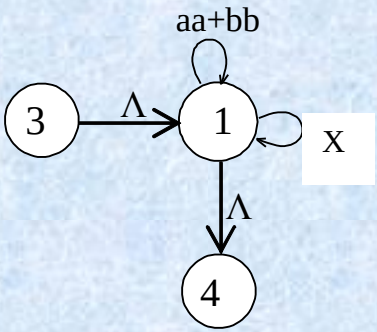
►Depends on the alphabet

►None of these

Question No: 10 (Marks: 1) - Please choose one



If above given TG is drawn like



Then what will be written in place of X.

► (ab+ba)(aa+bb)(ba+ab)

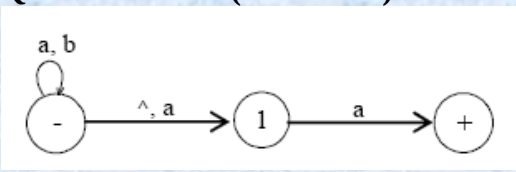
► (ab+ba)(aa+bb)(ab+ba)

► **(ab+ba)(aa+bb)*(ab+ba) (Page 31) rep**

► (ab+ba)(aa+bb)(ab+ba)*

ہر چیز کی ایک پہچان ہوتی ہے اور عقلمند کی پہچان غور و فکر کرنا ہے اور غور و فکر کی پہچان خاموشی ہے

Question No: 11 (Marks: 1) - Please choose one



Above given NFA-^ accepts____

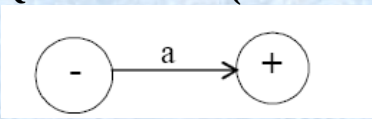
►bab

►a

►aba

►**a & aba rep**

Question No: 12 (Marks: 1) - Please choose one



Above given structure is a _____.

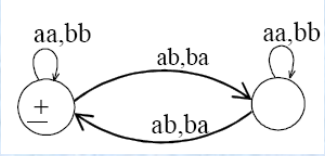
►FA

►TG

►FA & TG

►**NFA (Page 44)**

Question No: 13 (Marks: 1) - Please choose one



Above given TG has _____ RE.

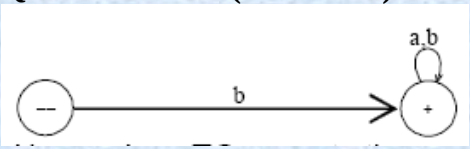
► (aa+aa+(ab+ab)(aa+ab)*(ab+ba))*

► **(aa+bb+(ab+ba)(aa+bb)*(ab+ba))* (Page 22)**

► (aa+bb+(ab+ba)(aa+bb)(ab+ba))*

►None of these

Question No: 14 (Marks: 1) - Please choose one



Above given TG has _____ RE.

►**b(a + b)* (Page 19)**

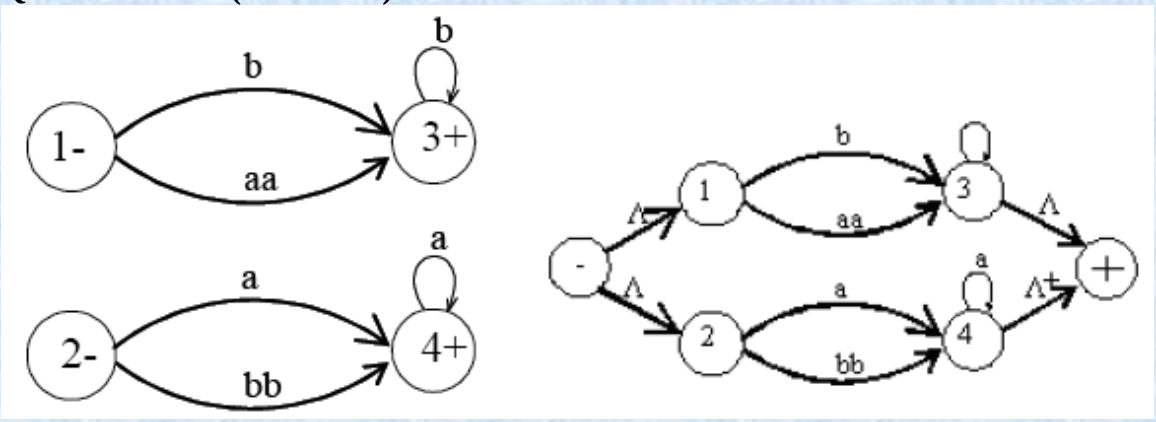
►b*(a + b)*

►b*(a + b)

►None of these

اپنی مرضی اور اللہ کی مرضی میں فرق کا نام غم ہے

Question No: 15 (Marks: 1) - Please choose one



Above given two TG's are _____.

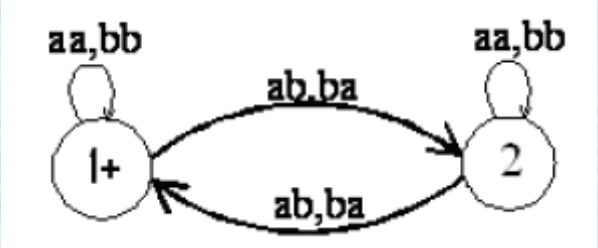
►**Equivalent (Page 26 - 27) rep**

►None-equivalent

►Not valid

►None of the given

Question No: 16 (Marks: 1) - Please choose one



Above given TG has _____ RE.

► (aa+bb+(ab+ba)(aa+bb)(ab+ba))*

► **(aa+bb+(ab+ba)(aa+bb)*(ab+ba))* (Page 22) rep**

► (aa+ba+(bb+ba)(ab+bb)(ab+aa))*

► (ab+ba+(ab+ba)(aa+bb)(ab+ba))*

وہ لوگ مبارک ہیں جو الفاظ سے نصیحت نہیں کرتے بلکہ عمل سے کرتے ہیں

MIDTERM EXAMINATION
Spring 2009
CS402- Theory of Automata (Session - 3)

Question No: 1 (Marks: 1) - Please choose one
S = {baa, ab}, then S* will not contain
►**abbaab**
►abab
►baabaa
►abbaa

Question No: 2 (Marks: 1) - Please choose one
$1^*(1 + \Lambda) = 1^*$ this statement is
►**True** **rep**
►False
►Sometimes true & sometimes false
►None of these

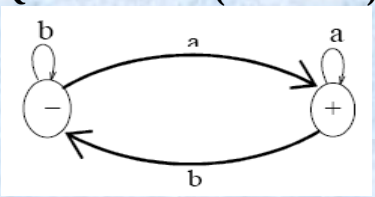
Question No: 3 (Marks: 1) - Please choose one
One language can be represented by more than one RE" this statement is____
►False
►**True** **(Page 9) rep**
►Can't be assumed
►None of these

Question No: 4 (Marks: 1) - Please choose one
a(a+b)*a+b(a+b)*b is RE for the language defined over S={a,b} having words beginning and ending with same letters
►**True** **(Page 14)**
►False
►Such a language is not regular
►None of these

Question No: 5 (Marks: 1) - Please choose one
If a language has RE, then that language can be expressed through TG.
►**True** **(Page 25) rep**
►False
►Depends on language
►None of these

کامیاب وکامران زندگی یہی ہے کہ جہاں رہو جس حال میں رہ خوش رہو

Question No: 6 (Marks: 1) - Please choose one



Above given FA can be expressed by

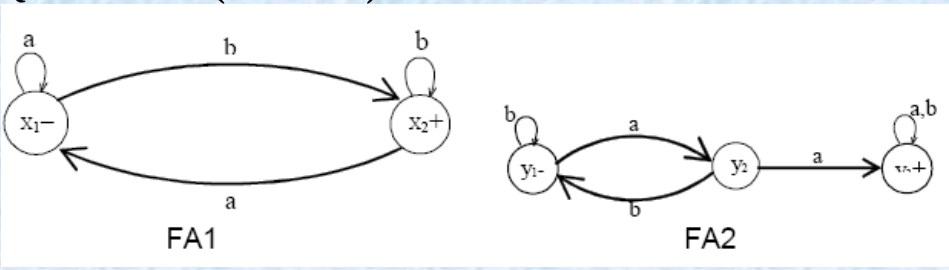
► **(a + b)*a** **(Page 14)**
► (a + b)*b
►a (a + b)*
►b (a + b)*

Question No: 7 (Marks: 1) Please choose one

In TG there may exist no paths for certain string.

►**True** **(Page 25) rep**
►False
►Depends on the language
►None of these

Question No: 8 (Marks: 1) - Please choose one



FA3 will express r1r2. then F3 will have _____ number of states in its diagram.

►8
►7
►**6** **(Page 36)**
►5

Question No: 9 (Marks: 1) - Please choose one

FA1 corresponds to r*, then FA1 must accept _____ string.

►**Every** **(Page 07)**
►Null
►Odd length
►Even length

اچھائی کرنے کے لئے ہمیشہ کسی بہانے کی تلاش میں رہو

Question No: 10 (Marks: 1) - Please choose one

In NFA, there may be more than one transition for certain letters and there may not be any transition for certain letters. This statement is _____.

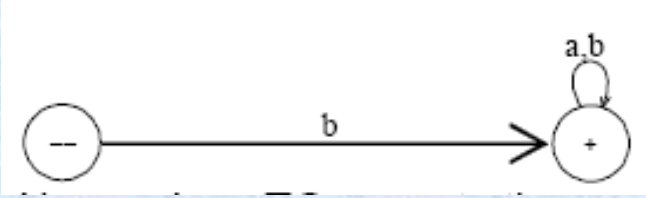
►False

►**True (Page 40)**

►Depends on language

►None of the given

Question No: 11 (Marks: 1) - Please choose one



Above given TG accepts the language in which all strings

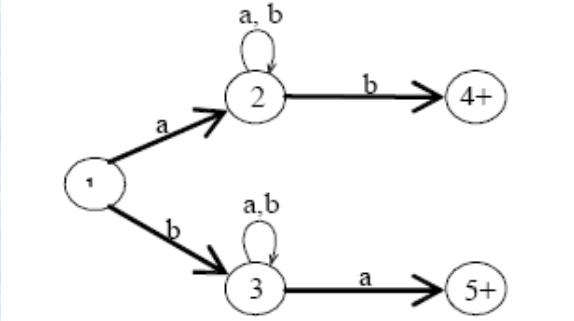
►Ends in b

►**Begins with b (Page 19)**

►Ends and begins with b

►None of the given

Question No: 12 (Marks: 1) - Please choose one



Above given TG represents the language____

►Begins and ends with same letters

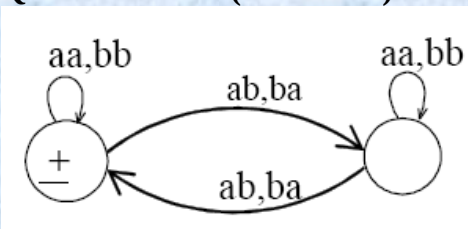
►**Begins and ends with different letters (Page 21) rep**

►Begins with a

►None of these

اطمینان قلب چاہتے ہو تو حسد سے دور رہو

Question No: 13 (Marks: 1) - Please choose one



Above given TG represents the language i.e.

►**EVEN-EVEN (Page 22)**

►PALINDROME

►FACTORIAL

►None of these

Question No: 14 (Marks: 1) - Please choose one

FA1 and FA2 are two FA's representing two languages. Then FA3, which is sum of FA1 and FA2, will accept the strings which are

►Accepted by FA1 AND FA2

►Accepted by FA1 OR FA2

►**Accepted by FA1 AND/OR FA2 (Page 32)**

►None of the given options

Ref:- language corresponding to r1+ r2 is the union of corresponding languages L1 and L2, consists of the strings belonging to L1or L2 or both

Question No: 15 (Marks: 1) - Please choose one

a (a + b)* is the RE of language defined over S = {a, b} having at least one a

►**True**

►False

►Such a language does not exist

►None of the given options

Question No: 16 (Marks: 1) - Please choose one

(a + b)* a is RE for the language defined over S={a,b} having words not ending in b

►**True**

►False

►Such a language is not regular

►None of the given options

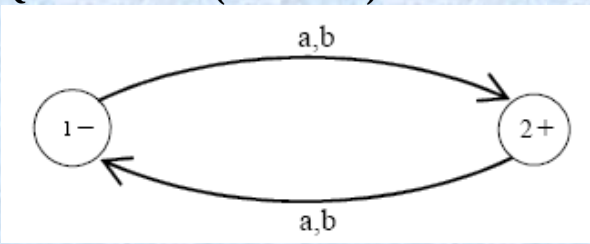
بہادر آدمی ایک مرتبہ جبکہ بزدل آدمی کئی مرتبہ مرتا ہے

MIDTERM EXAMINATION
Spring 2009
CS402- Theory of Automata (Session - 1)

Question No: 1 (Marks: 1) - Please choose one
Alphabet S = {a,bc,cc} has _____ number of letters.
►One
►Two
►**Three rep**
►Four

Question No: 2 (Marks: 1) - Please choose one
In which of the following language Rev(s)=s
►EQUAL
►INTEGER
►**PALINDROME (Page 6)**
►FACTORIAL

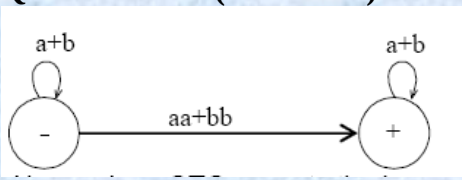
Question No: 3 (Marks: 1) - Please choose one
If S = {ab, bb}, then S* will not contain
►abbbab
►**bbba**
►bbbbab
►ababbb

Question No: 4 (Marks: 1) - Please choose one


Above given FA generates the language having strings of _____
►**ODD length**
►EVEN length
►Equal number of a's and b's
►None of these

پہلے وقتوں میں تعلیم کم تھی اور علم زیادہ، آج کل علم کم ہے اور تعلیم زیادہ

Question No: 5 (Marks: 1) - Please choose one



Above given GTG accepts the language in which strings

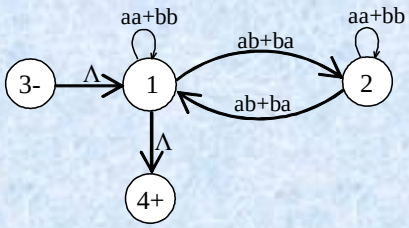
►**Contains double a or double b (Page 24) rep**

►Contains both a and double b

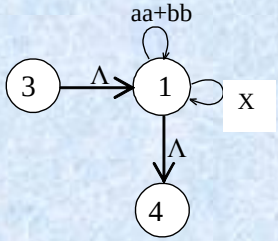
►Depends on the alphabet

►None of these

Question No: 6 (Marks: 1) - Please choose one



If above given TG is drawn like



Then what will be written in place of X.

► (ab+ba)(aa+bb)(ba+ab)

► (ab+ba)(aa+bb)(ab+ba)

► **(ab+ba)(aa+bb)*(ab+ba) (Page 31) rep**

► (ab+ba)(aa+bb)(ab+ba)*

Question No: 7 (Marks: 1) - Please choose one

FA3 expresses r1r2. Then initial state of FA3 will consist of

►Initial state of FA2

►**Initial state of FA1 (Page 35)**

►Initial states of both FA1 & FA2

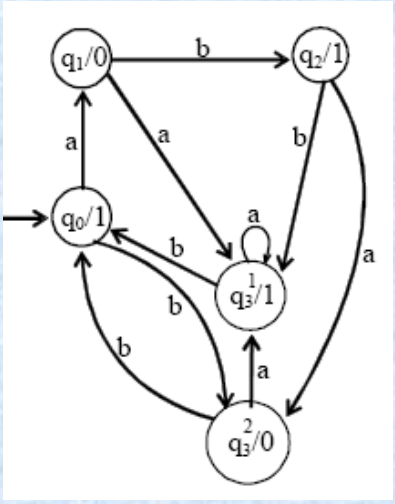
►Depends on FA's

ہر کسی کی روٹی نہ کھا بلکہ ہر شخص کو اپنی روٹی کھلا

Question No: 8 (Marks: 1) - Please choose one
FA3 expresses r1r2. Then there will be at least one final state of FA3 that consist of final state of FA1 and initial state of FA2.
►True
►**False (Page 35)**
►Depends on language
►None of these

Question No: 9 (Marks: 1) - Please choose one
Two machines are said to be equivalent if they print the same output string when the different input string is run on them
►True
►**False (Page 60)**
►Depends on language
►May be or may not be

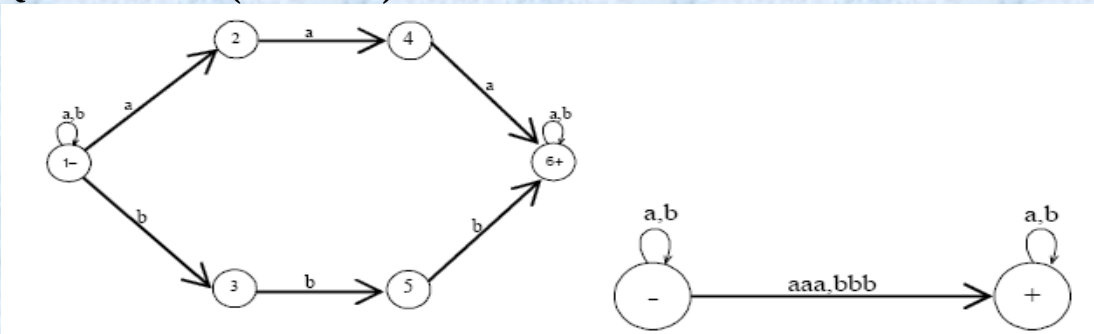
Question No: 10 (Marks: 1) - Please choose one
Running the string abbabbba on this Moore machine. The outputs will be_____



►**101111010 (Page 62)**
►01111010
►01011110
►01010101

فرقہ بندی ہی ہماری قوم کا زوال کا باعث ہے

Question No: 11 (Marks: 1) - Please choose one



Above given TG's are _____.

►None of these

►**Equivalent (Page 21) rep**

►Non-equivalent

►TG's are not valid

Question No: 12 (Marks: 1) - Please choose one

TG can have more than one initial state.

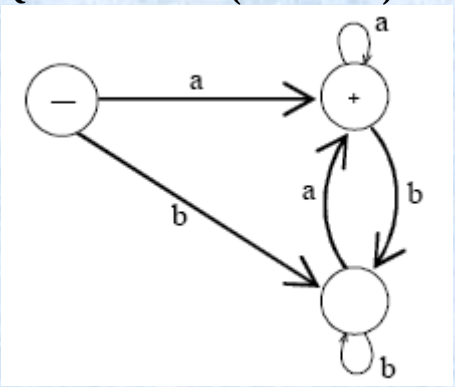
►**True (Page 18)**

►False

►Depends on alphabets

►None of these

Question No: 13 (Marks: 1) - Please choose one



Above given FA accepts null string.

►True

►**False**

►FA is not valid

►None of these

روح کی گہرائیوں کو پیش نظر رکھ کر بے غرض دوستی اختیار کرو

Question No: 14 (Marks: 1) - Please choose one
If in an NFA, $\wedge$ is allowed to be a label of an edge then that NFA is called _____.
►Will not remain NFA
►NFA with
►**NFA with null string (Page 42) rep**
►Either "NFA with null string" OR "NFA with "

Question No: 15 (Marks: 1) - Please choose one
One FA has n states and m letters in the alphabet. Then FA will have _____ number of transitions in the diagram.
► (n)+(m)
► **(m)(n) OR (n)(m) rep**
►None of the given options
► (m)-(n)

Question No: 16 (Marks: 1) - Please choose one
(a+b)*a(a+b)*b(a+b)* is the RE of language defined over S={a,b} having at least one a and one b
►**True**
►False
►Such a language does not exist
►None of the given options

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CS402-Theory Of Automata

FINAL TERM MCQS

Prepared by: JUNAID MALIK

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1.Which of the following steps replaces multiple incoming transition edges with a single one proving Kleene's theorem part II?

a. Step 1
b. Step 2
c. Step 3
d. Step 4

2. Let FA1 accepts many strings and FA2 accepts no strings, then FA1+FA2 will be equal to:

a.FA1

b.FA2

C.FA2-FA1

d.(FA2)*

3. Let L be the language of all strings, defined over ∑ =(0,1), ending in 10. Which of the following strings are indistinguishable w.r.t L with z being 0?

a.100.101

b.111.101

c. 110 .101

d.010.101

4. If r1=(aa+bb)and r2=(a+b) then the language (aa+bb)(a+b) will bhe generated by_____

a.(r1)(r2)

b.(r1 +r2)

c.(r2)(r1)

d.(r1)*

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5. Introducing a new start state in the case of multiple states is step no._____ of proving Kleen's theorem part II.

a.1

b.2

c.3

d.4

6. The language having even numbers of a's and even number of b's defined over S={a,b} is called _____

a. EVEN-EVEN

B.ODD-ODD

c. PALINDROME

d. FACTORIAL

7. In NFA having no transition at a certain state, FA can be built by introducing :

a.Empty state

b. Combination of states

c .initial state

d. final state

8. For every three regular expression R, S, and T. the language denoted by R(S U T) and (RS)U(RT) are the _____

a .same

b. Different

c. R(SUT)is greater

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d. none of the given options

9. Which of the following string belongs to the language of the regular expression (aa*b)?

a. baabab

b. abbbaa

c.aaaaaa

d.aabaab

10. If L1' and L2' are regular languages, then L1, L2 will be

a. non regular

b. may be regular

c. Regular

d. none of the mentioned

11. Suppose the language L1 has 2 L2 has 2 states. If we have a machine M that accepts L1∩L2. Then the total number of states in M is equal to_____

a. 2

b. 4

c. 6

d. 8

12. If L is a regular language, then (L')' U L will be:

a. L

b.

c.

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d.

13. In Mealy machine, the output depends on_____

a. Only present state

b. Present state and Present output

c. Nothing

d. Type of input

14. Strings x,y,z belongs to ∑* such that xz∈ L but yz∉ L where L ⊆∑* are:

a. indistinguishable

b. Undetermined

c. Both distinguishable and indistinguishable

d. Distinguishable

15. Melay machine to increase the output string in magnitude by 1 is called:

a. Complementing machine

b. Decrementing machine

c. Incrementing machine

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d. Converting machine

16. Suppose we have FA3 (which is to FA1+FA2) then the final state of FA3 will be declared final if:

a. it corresponds to final states of both FA1 and FA2

b. it corresponds to final states FA1 only

c. it corresponds to final states FA2 only

d. It corresponds to any of the final states of FA1 or FA2

17. If we have the finite language and the number of states in the FA is n then the maximum number of letter in each word of the language that will be accepted by the given FA will be :

a. 1

b. n+1

c. n-1

d. n

18. Which of the following state is introduced while developing NFA for the closure of FA?

a. An initial state which should be final as well

b. simply an initial AL-JUNAID INSTITUTE OF GROUP

c. Final states

d. An initial state with loop for all letters

19. Length of EVEN-EVEN language is_____

a. Odd

b. such language doesn't exist

c. Even

d. sometimes even & sometimes odd

20. If FA1 corresponds to (a+b)* then FA1 must accept_____ strings/strings.

a. No

b. Odd length

c. Every

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d. Even length

21. In FA, initial stage is represented by :

a. leaventh state empty

b. drawing a circle in that state

c. drawing '+'sign in transtate

d. Drawing an arrowhead before the state

22. Which one of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?

a. Moore machine

b. Finite state machine

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c. Mealy machine

d. Deterministic finite state machine

23. Length of machine "AbBAbcd" defined over ∑=(Ab,B,c,d) is_____.

a. three

b. four

c. five

d. six

24. An FA is a collection of:

a. Finite states , finite transition and finite input letters

b. infinite states , infinite transition and infinite input letters

c. Only finite states , finite transition

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d. Only infinite states , infinite transition

25. Given the language L={ab, aa, baa}, which of the following strings are in L*?

i) abaabaaabaa
ii) aaaabaaaa
iii) baaaaabaaaab
iv) baaaaabaa

a. 1,2 & 3

b. 2, 3&4

c. 1,2 & 4

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d. 1, 3&4

26. In the context to make NFA for the concentration of FA1 and FA2 (FA1 accepting null string) which of the following option is correct?

a. initial states in both FAs

b. FA2 having initial state only

c. FA2 having final state only

d. Final states in both FAs

27. Every ___ is a ___ as well, but the converse may not be true.

a. TG ,FA

b. TG, GTG

c. FA ,GTG

d. FA, TG

28. NFA with null string has ____ initial state(s).

a. One

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b. Two

c. Three

d. Many

29. While finding RE corresponding to a TG, we connect the new start state with the old start state by____ transition.

a. a

b. b

c. Null

d. Re

30. If S=(x). then S* will be_____.

a. {^,x,xxx,xxxx,xxxx….}

b. {^,x,xxx,xxxx,xxxxx….}

c. {^,x,xx,xxx,xxxx….}

d. {^,x,xxxx,xxxxx,xxxxxx….}

31. The minimum length of string (except null string) of a language that starts and ends in the same letter will be:

a.1 AL-JUNAID INSTITUTE OF GROUP

b. 2

c.3

d. 4

32. If S={ab, bb} , then S* will not contain_____.

a. abbbab

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b. bbbbab

c. ababbb

d. bbba

33. Which of the following machine has only one initial state?

a. Moore machine

b. Finite state machine

c. Deterministic finite state machine

d. Non deterministic finite state machine

34. Which of the following diagram is very rigid in order to express any language?

a. TG

b NFA

c GTG

d FA

35. Let L be the language of all strings, defined over ∑ =(0,1), ending in 111. Which of the following strings are indistinguishable w.r.t L with z being 11?

a 111, 101

b 100, 101

c110, 101

d 010, 101

36. Mealy machine can have _____ final states.

a. One

b More than one but finite

c More than one but infinite

d. Zero

37. In Moore machine , output is produced over the change of:

a. transition

b. transition and states

c. States

d. None of the mentioned

38. Lets we have two regular expressions R1 ={xx+yy}and R2={x+y} . Which one of the following is the correct regular expression for the union of R1 and R2?

a. (xx +yy) (x +y)

b. (xx +yy) +(x +y)*

c. (xx+ yy) +(x+ y)

d. ((xx +yy)+(x+ y))*

39. FA corresponding to an NFA can be built by introducing a state corresponding to the combination of states, for a letter having

a. no transition at certain state

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b. one transition at certain state

c. more than one transitions at certain state

d. none of the given options

40. The situation there is no way to leave after entry is called

a Davey John locker

b. initial state

c. finial state

d. non- final state

41. Which one of the following word is not accepted by the given regular expression?

a. aaabab

b. aaaababb

c. abbaab

d. aabbabb

42. According to the theory of automata, there are _____ types of languages.

a. one

b. Two

c. three

d. four

43. Which of the following word is not accepted by the given regular expression?

a. ababaaaab

b. bababbbba

c. baabaabba

d. abbaaabba

44. Regular languages are closed under the following operations.

a. Union only

b. Concatenation, closure only

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c. Union, Concatenation and closure

d. Regular languages are not closed under any operation.

45. There can be more than ____ FA for a certain language but for ____FA there is only one language associated with it.

a. one, one AL-JUNAID INSTITUTE OF GROUP

b. one, two

c. two, three

d. two ,one

46. There is no compulsion that each state must have an on outdoing edge for every input variable in:

a. Finite automata

b. Transition graph

c. Both finite Automata and Transition Graph

d. Transition Table

47. FA is also called_____.

a. TG

b. GTG

c. NFA

d. DFA

48. If r1 and r2 are regular expressions then (r1*r2) is_____

a. FA

b. RE (Regular expression

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c. TG

d. GTG

49. Which of the following is the minimal number of states for a finite automaton accepting the language of all strings defined over any alphabet set?

a. 1

b. 2

c. 3

d. 4

50. Keeping view language of all strings ending with ‘a’ for which symbol we will take a loop on the final state of its transition diagram?

a. A

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b. B

c. C

d. D

51. Which of the following statements is true about NFA with Null String?

a. Infinite state

b. Infinite set of letters

c. Infinite set of transitions

d. Transition of null string is allowed at any stage

52. Which one of the following diagrams expresses languages more simply?

a. FA

b. NFA

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c. TG

d. GTG

53. The language of all strings defined over alphabet set={a,b} that does not end with 'a' actually end with:

a. a and ^

b. b

c. b and^

d ^

54. Let S={aa,bb}, then S* will have the _____ string.

a. abba

b. aabbbaa

c. bbaab

d. ^

55. Formal is also known as_____.

a. Syntactic language

b. Semantic language

c. informal language

d. none of these

56. There may be more than one transition for a certain letter on a state in:

a. Non-Deterministic finite automata

b. finite automata

c. transition table

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d. Moore Machine

57. A loop at a state is supposed to be____ transition while converting Moore machine into an equivalent Maley machine.

a. incoming

b. outgoing

c. Both Incoming and incoming

d. Complex

58. FA of EVEN language shows null string when_____.

a. initial state is final as well AL-JUNAID INSTITUTE OF GROUP

b EVEN does not accept null

C One state is declared null

d. None of these

59. Which of the following statement is true about GTG?

a Transitions are based on input letters

b Transitions are based on specified substrings

c Transitions are based on regular expressions

d Transitions are based on alphabet set

60. Which of the following is the bypass and state elimination step in the context of Kleene's theorem part II proof?

a.1

b.2

c.3

d.4

61. Which of the following is free of non-determinism?

a. TG

b. NFA

c. NFA-^

d. FA

62. There is no question of accepting any language in:

a. FA

b. TG

c.GTG

d. Moore machine

63. A string will be accepted by an NFA if there exist_____one successful path.

a. Atmost

b. Maximum

c. atleast

d. None of these

64. Kleene's theorem part I expresses the relationship between_____

a. FA and TG

b. TG and RE

c. RE and FA

d. FAand RE

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65. Keeping in view the discussion by Martin, how any states are required to recognize the language of all strings of length 2 or more defined over ∑={a,b} with 'b' being the second letter from right?

a. 6

b. 7

c. 8

d. 9

66. FA and _____ are same excepts that _____has unique symbol for each transition.

a. FA, TG

b. NFA, TG

c. GTG, NFA

d. NFA,FA

67. Subtraction of binary number is possible through.

a. Both complementing and incrementing machine

b. incrementing machine

c. complementing machine

d. Converting machine

68. Which of the following is the bypass and state elimination step in the context of Kleene's theorem part II proof?

a. 1

b. 2

c. 3

d. 4

69. Keeping in view the discussion by Martin, how any states are required to recognize the language of all strings of length 2 or more defined over ∑={a,b} with 'b' being the second letter from right?

a.6

b. 7

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c. 8

d.9

70.let L be the language of all strings. Defined over ∑ (0,1) ,ending in 10.which of the following strings

Are indistinguishable with respect to L with z being 0

a.100.101

b.111.101

c. 110 .101

d.010.101

71.IF r1 =(aa +bb)and r2 =(a+ b) then the language(aa +bb)(a+ b) will be generated by _____

a.(r1)(r2)

b.(r1 +r2)

c.(r2)(r1)

d.(r1)*

72. Indroducing new start state in case of multiple start states in the step no_____

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a.1

b.2

c.3

d.4

73 THE language having even number of a ‘ s and even number of b’ s defined over S=(a, b) is called____

a. EVEN-EVEN

B.ODD-ODD

c. PALINDROME

d. FACTORIAL

74 In NFA having no transition at certain state .FA can be build by introducing.

A .Empty state

b. Combination of states

c .initial state

d. final state

75 .for every three regular expressions R,S. and T. the languages denoted by R(S U T)and (RS)U(RT)are the_____

a .same

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b. Different

c. R(SUT)is greater

d. none of the given options

76. Null strings can be specified on edges in:

a. Finite Automata

b Non-Deterministic Finite Automata

c. Melay Machine

d. Transition graph

77. What is false about the PLAINDROME LANGUAGE?

A Every word is reverse of itself.

B It is an infinite language.

C FA can be build for it.

D None of the given option

78. While finding RE corresponding to a TG, if TG has more than one start state then.

a Eliminate the old start state

b Replace the old start state with final state

c Replace the old final state with new start state

d. Introduce the new start state

79. All possible combination of string of a language including null string is referred as..

Kleene star closure of a languagec.

80.N! will be equal to:

c. n*(n-1)!cfd

81.Every NFA can be considered to be a____ as well, but the converse may not be true.

a. TG

b. FA

c. RE

d. GTG

82.In proving Kleene theorem II , if three statuses are connected then middle state is removed btconnecting first and third and writing corresponding RE in:

a.

b.

c. Concentration

d.

83.In _____ there must be transition for all the letters of the string.

a. NFA

b.GTG

c TG

d. FA AL-JUNAID INSTITUTE OF GROUP

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84.For a given Moore Machine , the input string is '101010' , thus the output string would be of length:
a. |Input|
b. |Input-1|
c. Cannot be predicted
d. Length of input string+1

85.The FA can be drawn for the regular expression (a+b)* with minimum_____ state(s).
a. 1
b.2
c. 3
d. 4

86.Which of the following does not contribute while finding out the length of strings?
^ ANSEWR

87.The language of all strings defined over alphabet set ={x,y} that ends with same letters will have the maximum length of:
a. Finate
b. non of these
c. Infinite
d. Equal

88.Considering FA1 and FA2 having 2 states each. Now FA1+FA2 can have maximum_____ number of states.
a. 2
b. 3
c. More than 3
d. Non of them

89.Which one of the following is RE for the language defined over ∑={a,b} having all the words starting with a?

a(a+b)*

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90.An _____ can be considered to be an intermediate structure between finite automaton and transition graph.
a. RE
b. NFA
c. GTG
d. None

91.In order to make NFA for the union of FA1 and FA2, the new initial state should be liked to:
a. initial state of FA1 only
b. initial and final states of FA1 and FA2 respectively
c. final and initial states of FA1 and FA2 respectively
d. Initial states of both FAs

92.We cannot construct an NFA for the language of _____ defined over alphabet set{a,b}.
a. Palindromes
b. Even
c. Odd
d. Integers

93.The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by the …… production trees
a. one
b. Two
c. More then one / Different
d. At most one

94.What do automata mean?
a. something done manually
b. something that works automatically / "something are done automatically
c. something done automatically

95.According to theory of automata there
a. 1
b. 2
c. 3

d. 4

96.The Minimum length of the string (except bull string) of a language that starts and ends in the same letters will be:

a.1

b. 2

c. 3

d. All of above

97. If S= {ab,bb} then S* will not contain_____

a. Babb

b. Bbb

c. Bbba

d abbb

98.Which of the following machines has only one initial state and no final state?

a. More machine

b. Finite state machine

c.Deterministic finite state machine

d.Non deterministic finite state machine

98.Which of the following machines has only one initial state and no final state?

a. More machine

b. Finite state machine

c.Deterministic finite state machine

d.Non deterministic finite state machine

99.Which of the following diagram is very rigid in order to express any language ?

a.TG

b.NFA-^

c.GTG

d.FA

100.If S={a},then S+will be_____
a.{a,aa ,aaa , aaaa…}
b.{a,aa,aaa,aaaa…}

c.{a,aaa,aaaaa,aaaaaaa,…..}

d. {aa, aaaa,aaaaaa,aaaaaaaa,…}

101.Let L be the language of the all string defined over over$\sum = \{0,1\}$ ending in10. Which of the following string are indistinguishable with respect to L with z being 11?
a.111,101
b.001,101
c.010,101
d.111,111
102.Melay machine can have _____final state
1) Zero
2) One
3) More than one but finite
4) More than one but infinite

103.Given the Language L = {ab, aa,baa},which of the following string are in L*?
5) Abaabaaaabaa
6) Aaaabaaaa
7) Baaaaabaaaaab
8) Baaaaabaa
a)1,2 and 3
b)2,3 and 4
c)1,2 and 4

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d)1,3 and 4

104.If L1 and L2 are regular Language L1∩ ($L2 \cup L1$) will be
a.non regular
b.may be regular
c. Regular
d.none of the mentioned
105.In Mealy Machine the out put depends on_____
a. state
b.previous state
c.state and input
d.only input
106.There is no question of accepting any language in.
a.Mealy machine
b. Moore Machine
c.Moore and mealy
d.none of these.ge
107.Thestate where there is no way to leave after entry is called___
a.Davey john Locker
b.Initial State

c.Final state

d.Non-final state

108.FA corresponding to an NFA can be built by introducing an empty state for a letter having
a.No transaction at certain state
b.More than one transition at certain state

c.None of the given option

109.Which of the following diagram express language more simply?
a.Tg
B.Gt
c.GTG
d.All of above
110.Automata is the plural of_____
a.Auto
b. Automation

c.Meta
d.All of above

111.If A and B are regular Language !(A∪B) is
a.Regular
b.none regular
c.may be regular
d.none of the mentioned
112.In NFA having no transition at certain state FA can be built by introducing.
a. Empty state.
B.Final state
C.initial state
D none of these

113.Consuming FA1 and FA2 having 2 STATES each. Now FA1+FA2 can have maximum_____number of state
a.2
b.3
c. More then 3
d. None of the given
114.In an FA when there is no path starting from initial state and ending state in final state then that FA_____
a.Reject all
b. Accept all string
c.Reject and accept all
d.None of above
115.According to theory of automation there are_____type if language
a. Two
b.Three
c.One
d.Four
116.In Moore machine if the length of input string is 9 then the length of output string will be.
a.20
b.60
c.10

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d.40

117.When ODD language is expressed by an FA,then it will have minimum _____ state

a. One

b TWO

c THREE

d FOUR

118.[(a+b)(a+b)]*.given RE contact generate the string_____

a. aaabbbbccc

b. Bbbbbb

c. aabbcc

d aaabcbbcbacc

119.Which of the following state is true about GTG?

a. Transaction are based on regular expression

120.Every_____is a _____ as will, but the converts may not be true.

a. FA ,TG

b. TG RE

c. GTG

d TG

121.Which of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?

a.TG andRE

b.REand FA

c.FA and RE

d. Mealy Machine

122.The recursive method for defining a language has ____ steps

a. two

b. Three

c. four

d. five

123.Consider the following RE.

.A(a+b)b*

All of the following word are accepted except_____

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A Aa

b.Aabb
c.Bbbba
d.None of these

124.Which of the following regular expression represented same language?

1. (a+ab)*
2. 2(ba+a)*
3. A*(aa*b)*
4. (a*b)*

a.2 and 1
b.3 and 2
c.1 and 2
d.3 and 4

125 For every there regular expression R,S and T the Language denoted by R(SUT) and (RS) U(RT) are the

a.Different
b. Same
c.Equal
d.All of above

126 Alphabet S={a,bc,cc} has____number of letters.

a. two
b. three
c. four
d. five

127 An____can be considered to be an intermediate structure between Finite automation and Transition Graph.

a.FA
b.RE
cBoth a and b
d.NFA

128 Two FAs are said to be equivalent if they_____

a.Simple langue
b.Accept different language
c. Accept same language

d. All of above

129 There may be more then one transition for a certain letter on a state in

a. Finite Automata

b.In finite automata

d. None of these

130 _____can also help in proving Kleene Theorem III.

A. RE

b.Rna

c.FA

d.NFA

131 Kleene Theorem Part II expression the relationship between____

a.TG and FA

b. TG and RE

c.TG and NFA

d.None of these

132 FA corresponding to an NFA can be built by introducing an empty state for a letter having.

a. No transition at certain state

b.Transition at certain state

c.Transition at final state

d. None of these

133 FA is also called

a.FA

b.RE

c. DFA

d.All of above

134 If two Res generated same language then these Res are called ____

a.Equivalent FA

b. Equivalent RE

c.Equivalent DFA

d.none of these

135 We cannot an NFA for the language of ____ defend over alphabet set{a,b}

Palindromes (**answer**)

136 Kleene Theorem Part II expression the relationship between____
a. RE and FA
b.DFA and RE
c.FA and DFA
d.None of these
137 Let FA3 an FA corresponding to FA1 FA2 then the initial state of FA3 must correspond to the initial state of
a.Dfa
b.FA2 only
c. FA1 only
ddnone of these
138 Every FA should be___
a.Simple
b.Accurate
c.Non-deterministic
d. Deterministic
139 The minimum length of string (except null string) of a language that starts and ends in the different letter will be:
a.2
b.3
c.4
d.1
140 Which of the following will be the final state of FA3 obtained from the union of FA1 and FA2?
a.Initial state
B.Final states of FA1 or FA2
c.Initial state of FA1 orFA 2
d.None of the given

141.In concatenation we accept the initial state of FA2 automatically after the final state of FA1 because of:
a.initial state

b.Final state

c.We need just one final state

d.We need just one initial state

142 Let FA1 accepts many strings and FA2 accepts none then FA1+FA2 will be equal to:

a. FA2-FA1 (ANSWER)

143 The language {a ab aba bab} is _____.

a.Irregular

b.Normal

c. Regular

d.All of above

144 Decomposing a string into its valid units is referred as:

a.Customizing

b. Tokenizing

c. Decomposing

d. Splitting

145 Let FA3 be an FA corresponding to FA1+FA2, then the initial state of FA3Must corresponds to the initial state of

a.NFA

b.DFA

c.RE

d.FA1 or FA2

146 If FA1 corresponds to (a+b)* then FA1 must accept _____ string/strings. Select correct option: No Oddlength

a.Some

b. EVERY

c.Just one

d.None of these

147 A regular language can be:

a. irregular

b. infinite

c. non-deterministic

d. None of the given options

148 N into its valid units is referred as:

d. Tokenizing

149 The strings of FA2 are accepted first before the strings of FA1

a.FA

b.NA

c. Palindromes

d.Non of these

150 There _____ a language for which only FA can be built but not the RE.

a.Is

b.Are

c.Be may

d.None of above

151 Kleene's Theorem part I expresses the relationship between ___

a.TG and Na

b.NA and FA

c.TG and TG

d. FA and TG

152 We can create an equivalent ___ for a language for which we create an ___

a.NA

b.NFA, FA

c.RNA

d.None of these


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1.Which of the following steps replaces multiple incoming transition edges with a single one proving Kleene's theorem part II?

a. Step 1
b. Step 2
c. Step 3
d. Step 4

2. Let FA1 accepts many strings and FA2 accepts no strings, then FA1+FA2 will be equal to:

a.FA1

b.FA2

C.FA2-FA1

d.(FA2)*

3. Let L be the language of all strings, defined over ∑ =(0,1), ending in 10. Which of the following strings are indistinguishable w.r.t L with z being 0?

a.100.101

b.111.101

c. 110 .101

d.010.101

4. If r1=(aa+bb)and r2=(a+b) then the language (aa+bb)(a+b) will bhe generated by_____

a.(r1)(r2)

b.(r1 +r2)

c.(r2)(r1)

d.(r1)*

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5. Introducing a new start state in the case of multiple states is step no._____ of proving Kleen's theorem part II.

a.1

b.2

c.3

d.4

6. The language having even numbers of a's and even number of b's defined over S={a,b} is called _____

a. EVEN-EVEN

B.ODD-ODD

c. PALINDROME

d. FACTORIAL

7. In NFA having no transition at a certain state, FA can be built by introducing :

a.Empty state

b. Combination of states

c .initial state

d. final state

8. For every three regular expression R, S, and T. the language denoted by R(S U T) and (RS)U(RT) are the _____

a .same

b. Different

c. R(SUT)is greater

d. none of the given options

9. Which of the following string belongs to the language of the regular expression (aa*b)?

a. baabab

b. abbbaa

c.aaaaaa

d.aabaab

10. If L1' and L2' are regular languages, then L1, L2 will be

a. non regular

b. may be regular

c. Regular

d. none of the mentioned

11. Suppose the language L1 has 2 L2 has 2 states. If we have a machine M that accepts L1∩L2. Then the total number of states in M is equal to_____

a. 2

b. 4

c. 6

d. 8

12. If L is a regular language, then (L')' U L will be:

a. L

b.

c.

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d.

13. In Mealy machine, the output depends on_____

a. Only present state

b. Present state and Present output

c. Nothing

d. Type of input

14. Strings x,y,z belongs to $\sum^*$ such that $xz \in L$ but $yz \notin L$ where $L \subseteq \sum^*$ are:

a. indistinguishable

b. Undetermined

c. Both distinguishable and indistinguishable

d. Distinguishable

15. Melay machine to increase the output string in magnitude by 1 is called:

a. Complementing machine

b. Decrementing machine

c. Incrementing machine

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d. Converting machine

16. Suppose we have FA3 (which is to FA1+FA2) then the final state of FA3 will be declared final if:

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a. it corresponds to final states of both FA1 and FA2

b. it corresponds to final states FA1 only

c. it corresponds to final states FA2 only

d. It corresponds to any of the final states of FA1 or FA2

17. If we have the finite language and the number of states in the FA is n then the maximum number of letter in each word of the language that will be accepted by the given FA will be :

a. 1

b. n+1

c. n-1

d. n

18. Which of the following state is introduced while developing NFA for the closure of FA?

a. An initial state which should be final as well

b. simply an initial AL-JUNAID INSTITUTE OF GROUP

c. Final states

d. An initial state with loop for all letters

19. Length of EVEN-EVEN language is_____

a. Odd

b. such language doesn't exist

c. Even

d. sometimes even & sometimes odd

20. If FA1 corresponds to (a+b)* then FA1 must accept_____ strings/strings.

a. No

b. Odd length

c. Every

d. Even length

21. In FA, initial stage is represented by :

a. leaventh state empty

b. drawing a circle in that state

c. drawing '+'sign in transtate

d. Drawing an arrowhead before the state

22. Which one of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?

a. Moore machine

b. Finite state machine

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c. Mealy machine

d. Deterministic finite state machine

23. Length of machine "AbBAbcd" defined over ∑=(Ab,B,c,d) is_____.

a. three

b. four

c. five

d. six

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24. An FA is a collection of:

a. Finite states , finite transition and finite input letters

b. infinite states , infinite transition and infinite input letters

c. Only finite states , finite transition

d. Only infinite states , infinite transition

25. Given the language L={ab, aa, baa}, which of the following strings are in L*?

i) abaabaaabaa
ii) aaaabaaaa
iii)baaaaabaaaab
iv)baaaaabaa

a. 1,2 & 3

b. 2, 3&4

c. 1,2 & 4

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d. 1, 3&4

26. In the context to make NFA for the concentration of FA1 and FA2 (FA1 accepting null string) which of the following option is correct?

a. initial states in both FAs

b. FA2 having initial state only

c. FA2 having final state only

d. Final states in both FAs

27. Every ___ is a ___ as well, but the converse may not be true.

a. TG ,FA

b. TG, GTG

c. FA ,GTG

d. FA, TG

28. NFA with null string has ____ initial state(s).

a. One

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b. Two

c. Three

d. Many

29. While finding RE corresponding to a TG, we connect the new start state with the old start state by____ transition.

a. a

b. b

c. Null

d. Re

30. If S=(x). then S* will be_____.

a. {^,x,xxx,xxxx,xxxx….}

b. {^,x,xxx,xxxx,xxxxx.…}

c. {^,x,xx,xxx,xxxx….}

d. {^,x,xxxx,xxxxx,xxxxxx.…}

31. The minimum length of string (except null string) of a language that starts and ends in the same letter will be:

a.1 AL-JUNAID INSTITUTE OF GROUP

b. 2

c.3

d. 4

32. If S={ab, bb} , then S* will not contain_____.

a. abbbab

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b. bbbbab

c. ababbb

d. bbba

33. Which of the following machine has only one initial state?

a. Moore machine

b. Finite state machine

c. Deterministic finite state machine

d. Non deterministic finite state machine

34. Which of the following diagram is very rigid in order to express any language?

a. TG

b NFA

c GTG

d FA

35. Let L be the language of all strings, defined over ∑ =(0,1), ending in 111. Which of the following strings are indistinguishable w.r.t L with z being 11?

a 111, 101

b 100, 101

c110, 101

d 010, 101

36. Mealy machine can have _____ final states.

a. One

b More than one but finite

c More than one but infinite

d. Zero

37. In Moore machine , output is produced over the change of:

a. transition

b. transition and states

c. States

d. None of the mentioned

38. Lets we have two regular expressions R1 ={xx+yy}and R2={x+y} . Which one of the following is the correct regular expression for the union of R1 and R2?

a. (xx +yy) (x +y)

b. (xx +yy) +(x +y)*

c. (xx+ yy) +(x+ y)

d. ((xx +yy)+(x+ y))*

39. FA corresponding to an NFA can be built by introducing a state corresponding to the combination of states, for a letter having

a. no transition at certain state AL-JUNAID INSTITUTE OF GROUP

b. one transition at certain state

c. more than one transitions at certain state

d. none of the given options

40. The situation there is no way to leave after entry is called

a Davey John locker

b. initial state

c. finial state

d. non- final state

41. Which one of the following word is not accepted by the given regular expression?

a. aaabab

b. aaaababb

c. abbaab

d. aabbabb

42. According to the theory of automata, there are _____ types of languages.

a. one

b. Two

c. three

d. four

43. Which of the following word is not accepted by the given regular expression?

a. ababaaaab

b. bababbbba

c. baabaabba

d. abbaaabba

44. Regular languages are closed under the following operations.

a. Union only

b. Concatenation, closure only

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c. Union, Concatenation and closure

d. Regular languages are not closed under any operation.

45. There can be more than ____ FA for a certain language but for ____FA there is only one language associated with it.

a. one, one AL-JUNAID INSTITUTE OF GROUP

b. one, two

c. two, three

d. two ,one

46. There is no compulsion that each state must have an on outdoing edge for every input variable in:

a. Finite automata

b. Transition graph

c. Both finite Automata and Transition Graph

d. Transition Table

47. FA is also called_____.

a. TG

b. GTG

c. NFA

d. DFA

48. If r1 and r2 are regular expressions then (r1*r2) is_____

a. FA

b. RE (Regular expression

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c. TG

d. GTG

49. Which of the following is the minimal number of states for a finite automaton accepting the language of all strings defined over any alphabet set?

a. 1

b. 2

c. 3

d. 4

50. Keeping view language of all strings ending with ‘a’ for which symbol we will take a loop on the final state of its transition diagram?

a. A

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b. B

c. C

d. D

51. Which of the following statements is true about NFA with Null String?

a. Infinite state

b. Infinite set of letters

c. Infinite set of transitions

d. Transition of null string is allowed at any stage

52. Which one of the following diagrams expresses languages more simply?

a. FA

b. NFA

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c. TG

d. GTG

53. The language of all strings defined over alphabet set={a,b} that does not end with 'a' actually end with:

a. a and ^

b. b

c. b and^

d ^

54. Let S={aa,bb}, then S* will have the _____ string.

a. abba

b. aabbbaa

c. bbaab

d. ^

55. Formal is also known as_____.

a. Syntactic language

b. Semantic language

c. informal language

d. none of these

56. There may be more than one transition for a certain letter on a state in:

a. Non-Deterministic finite automata

b. finite automata

c. transition table

d. Moore Machine

57. A loop at a state is supposed to be____ transition while converting Moore machine into an equivalent Maley machine.

a. incoming

b. outgoing

c. Both Incoming and incoming

d. Complex

58. FA of EVEN language shows null string when_____.

a. initial state is final as well AL-JUNAID INSTITUTE OF GROUP

b EVEN does not accept null

C One state is declared null

d. None of these

59. Which of the following statement is true about GTG?

a Transitions are based on input letters

b Transitions are based on specified substrings

c Transitions are based on regular expressions

d Transitions are based on alphabet set

60. Which of the following is the bypass and state elimination step in the context of Kleene's theorem part II proof?

a.1

b.2

c.3

d.4

61. Which of the following is free of non-determinism?

a. TG

b. NFA

c. NFA-^

d. FA

62. There is no question of accepting any language in:

a. FA

b. TG

c.GTG

d. Moore machine

63. A string will be accepted by an NFA if there exist_____one successful path.

a. Atmost

b. Maximum

c. atleast

d. None of these

64. Kleene's theorem part I expresses the relationship between_____

a. FA and TG

b. TG and RE

c. RE and FA

d. FAand RE

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65. Keeping in view the discussion by Martin, how any states are required to recognize the language of all strings of length 2 or more defined over ∑={a,b} with 'b' being the second letter from right?

a. 6

b. 7

c. 8

d. 9

66. FA and _____ are same excepts that _____has unique symbol for each transition.

a. FA, TG

b. NFA, TG

c. GTG, NFA

d. NFA,FA

67. Subtraction of binary number is possible through.

a. Both complementing and incrementing machine

b. incrementing machine

c. complementing machine

d. Converting machine

68. Which of the following is the bypass and state elimination step in the context of Kleene's theorem part II proof?

a. 1

b. 2

c. 3

d. 4

69. Keeping in view the discussion by Martin, how any states are required to recognize the language of all strings of length 2 or more defined over ∑={a,b} with 'b' being the second letter from right?

a.6

b. 7

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c. 8

d.9

70.let L be the language of all strings. Defined over ∑ (0,1) ,ending in 10.which of the following strings

Are indistinguishable with respect to L with z being 0

a.100.101

b.111.101

c. 110 .101

d.010.101

71.IF r1 =(aa +bb)and r2 =(a+ b) then the language(aa +bb)(a+ b) will be generated by _____

a.(r1)(r2)

b.(r1 +r2)

c.(r2)(r1)

d.(r1)*

72. Indroducing new start state in case of multiple start states in the step no_____

a.1

b.2

c.3

d.4

73 THE language having even number of a ‘ s and even number of b’ s defined over S=(a, b) is called____

a. EVEN-EVEN

B.ODD-ODD

c. PALINDROME

d. FACTORIAL

74 In NFA having no transition at certain state .FA can be build by introducing.

A .Empty state

b. Combination of states

c .initial state

d. final state

75 .for every three regular expressions R,S. and T. the languages denoted by R(S U T)and (RS)U(RT)are the_____

a .same

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b. Different

c. R(SUT)is greater

d. none of the given options

76. Null strings can be specified on edges in:

a. Finite Automata

b Non-Deterministic Finite Automata

c. Melay Machine

d. Transition graph

77. What is false about the PLAINDROME LANGUAGE?

A Every word is reverse of itself.

B It is an infinite language.

C FA can be build for it.

D None of the given option

78. While finding RE corresponding to a TG, if TG has more than one start state then.

a Eliminate the old start state

b Replace the old start state with final state

c Replace the old final state with new start state

d. Introduce the new start state

79. All possible combination of string of a language including null string is referred as..

Kleene star closure of a languagec.

80.N! will be equal to:

c. n*(n-1)!cfd

81.Every NFA can be considered to be a____ as well, but the converse may not be true.

a. TG

b. FA

c. RE

d. GTG

82.In proving Kleene theorem II , if three statuses are connected then middle state is removed btconnecting first and third and writing corresponding RE in:

a.

b.

c. Concentration

d.

83.In _____ there must be transition for all the letters of the string.

a. NFA

b.GTG

c TG

d. FA

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84.For a given Moore Machine , the input string is '101010' , thus the output string would be of length:
a. |Input|
b. |Input-1|
c. Cannot be predicted
d. Length of input string+1

85.The FA can be drawn for the regular expression (a+b)* with minimum_____ state(s).
a. 1
b.2
c. 3
d. 4

86.Which of the following does not contribute while finding out the length of strings?
^ ANSEWR

87.The language of all strings defined over alphabet set ={x,y} that ends with same letters will have the maximum length of:
a. Finate
b. non of these
c. Infinite
d. Equal

88.Considering FA1 and FA2 having 2 states each. Now FA1+FA2 can have maximum_____ number of states.
a. 2
b. 3
c. More than 3
d. Non of them

89.Which one of the following is RE for the language defined over ∑={a,b} having all the words starting with a?

a(a+b)*

90.An _____ can be considered to be an intermediate structure between finite automaton and transition graph.
a. RE
b. NFA
c. GTG
d. None

91.In order to make NFA for the union of FA1 and FA2, the new initial state should be liked to:
a. initial state of FA1 only
b. initial and final states of FA1 and FA2 respectively
c. final and initial states of FA1 and FA2 respectively
d. Initial states of both FAs

92.We cannot construct an NFA for the language of _____ defined over alphabet set{a,b}.
a. Palindromes
b. Even
c. Odd
d. Integers

93.The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by the …… production trees
a. one
b. Two
c. More then one / Different
d. At most one

94.What do automata mean?
a. something done manually
b. something that works automatically / "something are done automatically
c. something done automatically

95.According to theory of automata there
a. 1
b. 2
c. 3

d. 4

96.The Minimum length of the string (except bull string) of a language that starts and ends in the same letters will be:

a.1

b. 2

c. 3

d. All of above

97. If S= {ab,bb} then S* will not contain_____

a. Babb

b. Bbb

c. Bbba

d abbb

98.Which of the following machines has only one initial state and no final state?

a. More machine

b. Finite state machine

c.Deterministic finite state machine

d.Non deterministic finite state machine

98.Which of the following machines has only one initial state and no final state?

a. More machine

b. Finite state machine

c.Deterministic finite state machine

d.Non deterministic finite state machine

99.Which of the following diagram is very rigid in order to express any language ?

a.TG

b.NFA-^

c.GTG

d.FA

100.If S={a},then S+will be_____

a.{a,aa ,aaa , aaaa…}

b.{a,aa,aaa,aaaa…}

c.{a,aaa,aaaaa,aaaaaaa,…..}

d. {aa, aaaa,aaaaaa,aaaaaaaa,…}

101.Let L be the language of the all string defined over over$\sum = \{0,1\}$ ending in10. Which of the following string are indistinguishable with respect to L with z being 11?

a.111,101

b.001,101

c.010,101

d.111,111

102.Melay machine can have _____final state

1) Zero
2) One
3) More than one but finite
4) More than one but infinite

103.Given the Language L = {ab, aa,baa},which of the following string are in L*?

5) Abaabaaaabaa
6) Aaaabaaaa
7) Baaaaabaaaaab
8) Baaaaabaa

a)1,2 and 3

b)2,3 and 4

c)1,2 and 4

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d)1,3 and 4

104.If L1 and L2 are regular Language L1∩ ($L2 \cup L1$) will be
a.non regular
b.may be regular
c. Regular
d.none of the mentioned
105.In Mealy Machine the out put depends on_____
a. state
b.previous state
c.state and input
d.only input
106.There is no question of accepting any language in.
a.Mealy machine
b. Moore Machine
c.Moore and mealy
d.none of these.ge
107.Thestate where there is no way to leave after entry is called___
a.Davey john Locker
b.Initial State

c.Final state

d.Non-final state

108.FA corresponding to an NFA can be built by introducing an empty state for a letter having
a.No transaction at certain state
b.More than one transition at certain state

c.None of the given option

109.Which of the following diagram express language more simply?
a.Tg
B.Gt
c.GTG
d.All of above
110.Automata is the plural of_____
a.Auto
b. Automation

c.Meta
d.All of above

111.If A and B are regular Language !(AUB) is
a.Regular
b.none regular
c.may be regular
d.none of the mentioned
112.In NFA having no transition at certain state FA can be built by introducing.
a. Empty state.
B.Final state
C.initial state
D none of these

113.Consuming FA1 and FA2 having 2 STATES each. Now FA1+FA2 can have maximum_____number of state
a.2
b.3
c. More then 3
d. None of the given
114.In an FA when there is no path starting from initial state and ending state in final state then that FA_____
a.Reject all
b. Accept all string
c.Reject and accept all
d.None of above
115.According to theory of automation there are_____type if language
a. Two
b.Three
c.One
d.Four
116.In Moore machine if the length of input string is 9 then the length of output string will be.
a.20
b.60
c.10

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d.40

117.When ODD language is expressed by an FA,then it will have minimum _____ state

a. One
b TWO
c THREE
d FOUR

118.[(a+b)(a+b)]*.given RE contact generate the string_____

a. aaabbbbccc
b. Bbbbbb
c. aabbcc
d aaabcbbcbacc

119.Which of the following state is true about GTG?

a. Transaction are based on regular expression

120.Every_____is a _____ as will, but the converts may not be true.

a. FA ,TG
b. TG RE
c. GTG
d TG

121.Which of the following machine is represented as a pictorial representation with states and directed edges labeled by an input letter along with an output character?

a.TG andRE
b.REand FA
c.FA and RE
d. Mealy Machine

122.The recursive method for defining a language has ____ steps

a. two
b. Three
c. four
d. five

123.Consider the following RE.

.A(a+b)b*

All of the following word are accepted except_____

A Aa

b.Aabb

c.Bbbba

d.None of these

124.Which of the following regular expression represented same language?

1. (a+ab)*
2. 2(ba+a)*
3. A*(aa*b)*
4. (a*b)*

a.2 and 1

b.3 and 2

c.1 and 2

d.3 and 4

125 For every there regular expression R,S and T the Language denoted by R(SUT) and (RS) ∪(RT) are the

a.Different

b. Same

c.Equal

d.All of above

126 Alphabet S={a,bc,cc} has____number of letters.

a. two

b. three

c. four

d. five

127 An____can be considered to be an intermediate structure between Finite automation and Transition Graph.

a.FA

b.RE

cBoth a and b

d.NFA

128 Two FAs are said to be equivalent if they_____

a.Simple langue

b.Accept different language

c. Accept same language

d. All of above

129 There may be more then one transition for a certain letter on a state in

a. Finite Automata

b.In finite automata

d. None of these

130 _____can also help in proving Kleene Theorem III.

A. RE

b.Rna

c.FA

d.NFA

131 Kleene Theorem Part II expression the relationship between____

a.TG and FA

b. TG and RE

c.TG and NFA

d.None of these

132 FA corresponding to an NFA can be built by introducing an empty state for a letter having.

a. No transition at certain state

b.Transition at certain state

c.Transition at final state

d. None of these

133 FA is also called

a.FA

b.RE

c. DFA

d.All of above

134 If two Res generated same language then these Res are called ____

a.Equivalent FA

b. Equivalent RE

c.Equivalent DFA

d.none of these

135 We cannot an NFA for the language of ____ defend over alphabet set{a,b}

Palindromes (**answer**)

136 Kleene Theorem Part II expression the relationship between____

a. RE and FA

b.DFA and RE

c.FA and DFA

d.None of these

137 Let FA3 an FA corresponding to FA1 FA2 then the initial state of FA3 must correspond to the initial state of

a.Dfa

b.FA2 only

c. FA1 only

ddnone of these

138 Every FA should be___

a.Simple

b.Accurate

c.Non-deterministic

d. Deterministic

139 The minimum length of string (except null string) of a language that starts and ends in the different letter will be:

a.2

b.3

c.4

d.1

140 Which of the following will be the final state of FA3 obtained from the union of FA1 and FA2?

a.Initial state

B.Final states of FA1 or FA2

c.Initial state of FA1 orFA 2

d.None of the given

141.In concatenation we accept the initial state of FA2 automatically after the final state of FA1 because of:

a.initial state

b.Final state

c.We need just one final state

d.We need just one initial state

142 Let FA1 accepts many strings and FA2 accepts none then FA1+FA2 will be equal to:

a. FA2-FA1 (ANSWER)

143 The language {a ab aba bab} is _____ .

a.Irregular

b.Normal

c. Regular

d.All of above

144 Decomposing a string into its valid units is referred as:

a.Customizing

b. Tokenizing

c. Decomposing

d. Splitting

145 Let FA3 be an FA corresponding to FA1+FA2, then the initial state of FA3Must corresponds to the initial state of

a.NFA

b.DFA

c.RE

d.FA1 or FA2

146 If FA1 corresponds to (a+b)* then FA1 must accept _____ string/strings. Select correct option: No Oddlength

a.Some

b. EVERY

c.Just one

d.None of these

147 A regular language can be:

a. irregular

b. infinite

c. non-deterministic

d. None of the given options

148 N into its valid units is referred as:

d. Tokenizing

149 The strings of FA2 are accepted first before the strings of FA1

a.FA

b.NA

c. Palindromes

d.Non of these

150 There _____ a language for which only FA can be built but not the RE.

a.Is

b.Are

c.Be may

d.None of above

151 Kleene's Theorem part I expresses the relationship between ___

a.TG and Na

b.NA and FA

c.TG and TG

d. FA and TG

152 We can create an equivalent ___ for a language for which we create an ___

a.NA

b.NFA, FA

c.RNA

d.None of these

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